

Task 1 – Understanding the Industry Problem

1.1 Industry Background

Motor insurance is a major part of the insurance industry that provides financial protection against losses associated with motor vehicles and their use. In return for an insurance premium, an insurer assumes certain financial risks on behalf of the policyholder according to the terms and conditions of the insurance policy.

For an insurance company, managing risk is a fundamental part of the business. The insurer receives premiums from a large number of policyholders, while only a proportion of those policyholders are expected to make claims during a given period. The insurer therefore needs to understand the characteristics associated with insurance claims in order to manage its overall portfolio of risk effectively.

Motor insurance risk can be considered from multiple perspectives. Two particularly important aspects are **claim frequency** and **claim severity**. Claim frequency concerns how often claims occur, while claim severity concerns the financial cost associated with claims. Understanding both is important for effective insurance risk management and pricing.

This project focuses specifically on **motor insurance claim frequency**.

1.2 Organizational Context

The organization considered in this consultancy project is a hypothetical motor insurance company that manages a portfolio of motor insurance policies.

The company possesses historical information about its policyholders, vehicles, geographical characteristics, insurance exposure, and historical claims. However, historical data alone does not automatically provide useful business knowledge. Statistical analysis is required to identify relationships within the data and determine which characteristics are associated with differences in claim frequency.

The organization therefore requires a data-driven approach that can transform historical insurance data into information that can support risk management and insurance-related decision-making.

The consultancy team will act as an independent statistical consulting company responsible for analysing the organization's historical insurance data and providing evidence-based recommendations.

1.3 Business Problem

The central business problem is the difficulty of accurately understanding and assessing differences in **motor insurance claim frequency** across policyholders.

Policyholders are not necessarily exposed to the same level of risk. Their characteristics, vehicle characteristics, geographical factors, and length of exposure to insurance may differ. These differences may be associated with different patterns of claim occurrence.

The organization therefore needs to answer questions such as:

- Which policyholder and vehicle characteristics are associated with claim frequency?

- Are there statistically significant differences in claim frequency between different groups?
- Which statistical modelling approach is most appropriate for modelling claim frequency?
- How accurately can expected claim frequency be estimated?
- How can the resulting statistical evidence support insurance risk management and pricing-related decisions?

The freMTPL2freq dataset provides historical motor insurance observations that can be used to investigate these questions.

The statistical nature of the problem is particularly important because the response variable represents the **number of claims**, making claim frequency a count-based outcome rather than a conventional continuous measurement.

1.4 Importance of Solving the Problem

Effective risk assessment is important to an insurance organization because poor understanding of risk can negatively affect business decision-making.

If the risk associated with policies is underestimated, an insurer may take on more risk than expected. Conversely, if risk is substantially overestimated, the organization may make inefficient or less competitive decisions.

A better understanding of claim frequency can therefore help the organization:

- identify characteristics associated with higher or lower claim frequency;
- understand the distribution of risk within its insurance portfolio;
- improve evidence-based risk segmentation;
- provide statistical evidence to support pricing-related decisions;
- improve risk monitoring and management;
- identify areas where further data collection or analysis may be valuable.

However, the statistical model should be considered a **decision-support tool rather than a replacement for professional insurance judgement**. Claim frequency represents only one component of insurance risk, and a complete insurance pricing decision would require additional information beyond the scope of this dataset.

1.5 Stakeholders

Several stakeholders may be affected by the analysis.

Insurance Management

Senior management requires information about the overall risk profile of the organization's portfolio. The results can support strategic decisions concerning risk management and the organization's insurance business.

Underwriting Teams

Underwriters assess insurance risks when deciding whether and under what conditions policies should be accepted. Statistical evidence about characteristics associated with claim frequency may provide additional information for this process.

Actuarial and Pricing Teams

Actuarial and pricing professionals develop statistical models to understand insurance risk and support pricing decisions. Claim-frequency modelling can provide an important component of this analytical process.

Policyholders

Customers may be affected by decisions relating to risk classification and insurance pricing. Therefore, the use of statistical models should consider fairness, transparency, and the potential consequences of risk-based decisions.

Regulators

Insurance organizations operate within regulatory environments. Regulatory stakeholders may have an interest in whether insurance decisions are appropriately justified, responsible, and compliant with relevant requirements.

1.6 Business Decisions Influenced by the Problem

The results of this consultancy project could support several organizational decisions.

Risk Segmentation

The organization could use statistical evidence to identify groups of policies associated with different levels of expected claim frequency.

Pricing Support

Claim-frequency estimates could provide one source of information for insurance pricing analysis. They would not, by themselves, determine the final premium because other components of insurance risk and business considerations would also need to be considered.

Portfolio Risk Management

The organization could use the analysis to understand how claim risk is distributed across its portfolio and identify concentrations of potentially higher-risk policies.

Underwriting Support

The findings could provide underwriters with additional evidence about characteristics associated with claim frequency when evaluating policies.

Risk Monitoring

The statistical framework could potentially be used as part of a wider monitoring system to identify changes in the risk profile of the organization's portfolio.

1.7 Consultancy Objectives

The primary objective of this consultancy project is:

To investigate the factors associated with motor insurance claim frequency and evaluate appropriate statistical modelling approaches to develop an evidence-based risk analytics framework that can support insurance risk management and pricing-related decision-making.

The specific objectives are to:

1. Investigate the characteristics of the motor insurance portfolio using descriptive statistical analysis.
2. Identify relationships between policyholder, vehicle, geographical, and exposure-related characteristics and claim frequency.
3. Apply appropriate statistical inference techniques to determine whether observed differences and relationships are statistically supported.
4. Develop and compare suitable statistical models for motor insurance claim frequency.
5. Evaluate the assumptions, strengths, limitations, and practical suitability of the selected models.
6. Critically evaluate the potential applicability of alternative statistical approaches, including PCA, Bayesian methods, experimental design, and time-series analysis.
7. Translate statistical findings into practical insights for insurance risk management and pricing-related decision-making.
8. Propose an innovative data-driven risk analytics and decision-support framework for the organization.
9. Develop recommendations based on statistical evidence, academic research, and industry feedback.

1.8 Scope of the Consultancy

This project focuses on **motor insurance claim frequency** using the selected historical dataset.

The analysis will investigate the relationship between available policy and vehicle characteristics and the observed number of claims. Statistical modelling will be used to estimate and understand claim frequency.

The project does **not** attempt to construct a complete commercial insurance pricing system. In particular, claim severity, operational costs, profit requirements, regulatory requirements, and other commercial factors are outside the primary scope of the claim-frequency analysis.

The proposed solution should therefore be interpreted as a **risk analytics and decision-support framework**, rather than an autonomous system for determining insurance premiums.

Task 2 – Research Landscape

2.1 Introduction

Motor insurance claim-frequency modelling is an established area of actuarial and statistical research. Insurance organizations need to estimate the expected frequency of claims in order to understand the risk profile of their portfolios and support decisions relating to underwriting, pricing, and risk management.

The statistical problem is different from conventional regression problems because the response variable represents the number of claims made by an insured policyholder during a period of exposure. Consequently, count-data models such as Poisson and Negative Binomial regression have traditionally played an important role in insurance claim-frequency modelling. More recently, researchers have investigated machine-learning, Bayesian, zero-inflated, and dynamic modelling approaches to address limitations associated with traditional statistical models.

The literature reviewed for this consultancy therefore focuses on the progression from traditional actuarial models towards more flexible statistical and machine-learning approaches, while considering the trade-off between predictive performance, interpretability, statistical assumptions, and practical usefulness.

2.2 Current Industry and Statistical Practice

A traditional approach to non-life insurance pricing is to model **claim frequency** and **claim severity** separately.

Claim frequency represents how often claims occur, while claim severity represents the financial cost associated with a claim. These components can subsequently contribute to an estimate of expected insurance losses.

For the present consultancy, the focus is specifically on claim frequency because the selected `freMTPL2freq` dataset provides information suitable for analysing the number of claims and the exposure period associated with policies.

Generalized Linear Models (GLMs) are particularly important in insurance because they allow non-Gaussian response variables to be modelled while providing interpretable relationships between risk factors and expected claims. Poisson regression is a natural starting point for claim counts, while Negative Binomial models provide an alternative when the variability in the observed claim counts exceeds that expected under a Poisson distribution.

Research comparing Poisson and Negative Binomial approaches demonstrates that both models can be useful for motor-insurance claim-frequency modelling, with the appropriate choice depending on the characteristics of the underlying data.

Therefore, the literature supports the use of a statistical diagnostic process rather than assuming that one distribution is universally appropriate.

2.3 Poisson and Negative Binomial Models

The Poisson model is one of the fundamental approaches for modelling insurance claim counts. Its attractiveness comes from its natural interpretation for count outcomes and its relatively straightforward integration into a GLM framework.

However, the Poisson distribution imposes a restrictive relationship between its mean and variance. In practical insurance data, claim counts may exhibit greater variability than the Poisson model permits. This phenomenon is commonly referred to as **overdispersion**.

The Negative Binomial model provides an additional dispersion parameter and can therefore accommodate greater variability in claim counts.

Studies of motor-insurance data have compared these approaches directly. For example, research applying Poisson and Negative Binomial models to automobile insurance data demonstrates their use for estimating claim frequency and supporting premium-related analysis.

This leads to an important methodological principle for the present project:

The Poisson GLM should be treated as an appropriate baseline rather than automatically assumed to be the final model.

After fitting the baseline model, its assumptions and dispersion characteristics should be examined. If the data demonstrate substantial overdispersion, a Negative Binomial model or another appropriate count model should be considered.

2.4 Excess Zeros and Zero-Inflated Models

Motor insurance portfolios commonly contain a large number of policyholders who do not make a claim during the observation period. Consequently, the distribution of claim counts may contain a very large proportion of zero observations.

This creates another modelling consideration.

Zero-inflated models distinguish between the process generating structural or excess zeros and the process generating the count of claims. Research using Malaysian motor-insurance data has compared Zero-Inflated Poisson (ZIP) and Zero-Inflated Negative Binomial (ZINB) models using both frequentist and Bayesian approaches. The study found that ZIP provided a better fit than ZINB for its particular data.

This finding is important because it demonstrates that model selection should be **data-dependent**. A zero-inflated model should not be introduced simply because the dataset contains many zero claims. The observed distribution and the underlying assumptions must first justify such an approach.

The literature therefore motivates us to examine the proportion and structure of zero claims during our descriptive and diagnostic analysis.

2.5 Limitations of Traditional GLMs

Traditional GLMs provide several important advantages for insurance applications.

Interpretability

The effect of individual rating variables can be examined and communicated to decision-makers.

Statistical foundation

GLMs provide a well-established framework for inference, model diagnostics, and hypothesis testing.

Practical applicability

Their structure is familiar within actuarial and insurance modelling environments.

However, traditional GLMs also have limitations.

A conventional GLM imposes a particular functional relationship between predictors and the response. Complex nonlinear relationships and interactions may therefore be difficult to capture unless additional transformations or interaction terms are explicitly introduced.

Clemente, Guerreiro and Bravo highlight this issue when comparing traditional GLMs with Gradient Boosting for motor-insurance frequency and severity modelling. Their research identifies the ability of Gradient Boosting

to capture nonlinear relationships and complex interactions as an important advantage over standard GLM specifications.

Thus, while GLMs provide an important interpretable baseline, they may not capture all of the complexity present in real insurance data.

2.6 Machine-Learning Approaches

The increasing availability of large insurance datasets has encouraged researchers to investigate machine-learning methods for risk modelling.

Tree-based ensemble methods such as Gradient Boosting can model nonlinear relationships and interactions without requiring these relationships to be explicitly specified in advance.

Clemente et al. compared Gradient Boosting with standard GLMs for motor-insurance claim frequency and severity. Their results demonstrated improved out-of-sample performance for Gradient Boosting in the frequency modelling component, illustrating the potential predictive advantage of more flexible approaches.

However, the result should not be interpreted as evidence that machine learning universally replaces GLMs.

A machine-learning model may provide improved predictive performance while being more difficult to interpret and communicate. This is particularly important in insurance because model outputs can influence decisions affecting customers.

Consequently, the literature suggests that the relevant question is not simply:

"Which model produces the highest predictive accuracy?"

Instead, an insurance organization should consider:

- predictive performance;
- calibration;
- interpretability;
- stability;
- statistical assumptions;
- implementation complexity;
- fairness;
- and business usefulness.

This aligns closely with the consultancy requirements of the present assignment, which explicitly states that prediction accuracy alone should not determine the quality of the modelling analysis.

2.7 Bayesian Approaches

Bayesian methods provide another direction for insurance claim-frequency modelling.

Unlike a purely frequentist approach, Bayesian modelling incorporates prior information and produces a posterior distribution for model parameters. This can be useful when uncertainty needs to be explicitly represented or when previous actuarial knowledge can be incorporated into the analysis.

Zhang et al. developed Bayesian CART models for insurance claim-frequency modelling and investigated Poisson, Negative Binomial, and Zero-Inflated Poisson distributions within a Bayesian tree framework. Their work highlights the potential of Bayesian approaches to combine flexible modelling with considerations of prediction and interpretability.

Other research has investigated Bayesian nonparametric regression for insurance claims, including applications to French motor-insurance data. Such approaches attempt to provide greater flexibility than conventional parametric regression models.

However, Bayesian approaches also introduce additional modelling decisions, including prior specification and computational complexity. Therefore, Bayesian modelling should be evaluated based on whether its additional flexibility and uncertainty representation provide meaningful advantages for the organization's decision problem.

This provides the basis for the later critical evaluation required by Task 8.

2.8 Dynamic and Time-Dependent Modelling

Traditional claim-frequency models often treat the relationship between risk factors and claims as relatively static. However, an insurer's risk environment can change over time due to changes in the insured population, vehicle technology, regulations, economic conditions, and driving behaviour.

Jiang, Li and Yang proposed a Dynamic Poisson State Space model for automobile claim-frequency prediction. Their model allows parameters to be updated as new claim information becomes available and was evaluated using six years of automobile insurance data. The study reported improved predictive performance compared with competing approaches in its empirical application.

This research demonstrates that claim-frequency modelling can potentially move beyond static prediction towards **continuously updated risk monitoring**.

The present dataset, however, does not provide the same type of longitudinal structure required for a full dynamic analysis. Therefore, time-series modelling will be treated as a potential future extension rather than being artificially applied to the current dataset.

This approach is consistent with the assignment, which asks the consultancy to critically discuss how time-series methods could be incorporated if historical time-dependent data were available.

2.9 Interpretability, Fairness and Responsible Insurance Modelling

Predictive performance is not the only consideration when statistical models are used in insurance.

Insurance pricing and risk classification can have direct financial consequences for policyholders. Therefore, transparency, explainability, and fairness are important considerations when selecting and implementing analytical models.

Recent research on motor-insurance pricing emphasizes that insurers must consider ethical and regulatory concerns alongside predictive modelling performance. The research investigates ways of identifying and mitigating biases in motor-insurance pricing and highlights the importance of measuring potential biases rather than treating predictive accuracy as the only objective.

This becomes particularly relevant if more complex machine-learning models are considered. A model may identify highly predictive variables or interactions that are difficult to justify from a business or ethical perspective.

Consequently, our consultancy will consider interpretability and responsible use of risk factors when evaluating alternative modelling approaches.

2.10 Critical Comparison of Existing Approaches

The literature indicates that no single modelling technique is universally optimal for motor-insurance claim frequency.

Approach	Main strengths	Main weaknesses	Relevance to our project
Poisson GLM	Interpretable, established, appropriate for count data	Restrictive variance assumption; may suffer from overdispersion	Essential baseline
Negative Binomial	Handles overdispersion	More complex than Poisson; may still fail to capture other distributional features	Important alternative
ZIP/ZINB	Can account for excess zeros	More assumptions and complexity; unnecessary if zero structure does not justify it	Conditional alternative
Gradient Boosting	Captures nonlinearities and interactions; strong predictive performance	Less transparent; greater complexity	ML comparison
Bayesian Regression/CART	Represents uncertainty; can incorporate prior information; flexible	More complex; prior specification and computation required	Required evaluation
Dynamic Poisson models	Can update risk relationships over time	Requires suitable longitudinal/time-dependent data	Future extension

The research therefore suggests that the most defensible strategy is **comparative modelling**, beginning with an interpretable statistical baseline and then evaluating whether more complex approaches provide meaningful additional value.

2.11 Research Gaps

Several gaps and opportunities emerge from the literature.

Gap 1 – Prediction versus interpretability

Machine-learning approaches may improve predictive performance, but increased complexity can reduce interpretability. The practical value of a model therefore depends on more than predictive accuracy.

Gap 2 – Data-dependent model selection

The presence of overdispersion or excess zeros means that Poisson models cannot automatically be assumed to be appropriate. Different datasets can favour different count distributions.

Gap 3 – Static versus dynamic risk

Many conventional approaches provide a relatively static representation of risk. Dynamic modelling provides an opportunity to update risk estimates as new information becomes available, although this requires appropriate longitudinal data.

Gap 4 – Responsible risk modelling

Increasing use of predictive analytics creates a need to consider fairness, transparency, and explainability alongside statistical performance.

Gap 5 – Practical decision support

Much of the research focuses on comparing statistical or machine-learning models. A further practical challenge is translating those model outputs into a decision-support framework that can be understood and used by insurance managers, underwriters, and other stakeholders.

2.12 Justification for the Proposed Analytical Approach

Based on the research landscape, the consultancy will adopt a **progressive and evidence-based modelling strategy**.

First, descriptive analysis will be used to understand the distribution of claim counts, exposure, policy characteristics, and potential risk factors.

Second, an interpretable **Poisson GLM** will provide a statistical baseline for claim-frequency modelling.

Third, the assumptions and diagnostic results of the Poisson model will be examined. In particular, dispersion and the distribution of zero claims will be investigated.

If the evidence indicates that the Poisson assumptions are inadequate, alternative count models such as **Negative Binomial** or, where justified, **zero-inflated models** will be considered.

Machine-learning approaches such as Gradient Boosting may then be evaluated as flexible alternatives capable of modelling nonlinear relationships and interactions.

The models will not be compared solely according to predictive accuracy. Evaluation will also consider:

- statistical validity;
- model assumptions;

- interpretability;
- calibration;
- complexity;
- limitations;
- and practical usefulness for insurance decision-making.

Bayesian methods will subsequently be critically evaluated according to their suitability for representing uncertainty and supporting decision-making. PCA and experimental designs will also be considered in accordance with the assignment requirements, but will only be recommended if they provide a meaningful benefit to the insurance problem.

Finally, potential time-series approaches will be discussed as a future extension if the organization obtains suitable time-dependent claims data.

This approach allows the consultancy to balance **traditional actuarial interpretability with modern predictive modelling**, while maintaining a clear connection between statistical analysis and organizational decision-making.

2.13 Summary

The research landscape demonstrates an evolution in motor-insurance claim-frequency modelling from traditional Poisson and Negative Binomial GLMs towards more flexible zero-inflated, Bayesian, machine-learning, and dynamic approaches.

Traditional GLMs remain valuable because of their interpretability, statistical foundation, and practical relevance. However, their assumptions may limit their ability to represent overdispersion, excess zeros, nonlinear relationships, and complex interactions. Machine-learning approaches can address some of these limitations but introduce challenges relating to interpretability, complexity, and responsible use.

The literature therefore supports a comparative modelling strategy rather than the selection of a single technique in advance. For this consultancy, the proposed approach is to begin with an interpretable statistical baseline, test its assumptions, evaluate appropriate alternative models, and ultimately recommend the approach that provides the best balance between statistical evidence, predictive performance, interpretability, and organizational usefulness.

This research landscape provides the foundation for the subsequent dataset analysis and statistical modelling stages of the consultancy project.

Task 3 – Dataset Understanding and Descriptive Analysis

This task examines the selected freMTPL2freq motor-insurance dataset to understand the structure of the portfolio, assess data quality, describe the main variables, and identify initial patterns in observed claim frequency. The analysis is descriptive only; statistical significance and predictive modelling are considered in later tasks.

Dataset Description

The dataset contains 678,013 motor-insurance policy observations and 12 variables. The variables cover claim counts, insurance exposure, driver characteristics, vehicle characteristics and geographical information. The response of interest is ClaimNb, which records the number of claims associated with a policy observation, while Exposure records the corresponding insurance exposure.

The dataset is suitable for the consultancy objective because it allows claim occurrence to be compared across policyholder, vehicle and geographical characteristics while taking differences in exposure into account.

Variable Description

Table 3.1 summarises the variables used in the descriptive analysis. IDpol is treated as an identifier rather than as a numerical risk factor.

Variable	Type	Role / description
IDpol	Identifier	Policy identifier; not treated as a numerical predictor.
ClaimNb	Count	Number of claims recorded for the policy observation.
Exposure	Numerical	Insurance exposure associated with the observation.
Area	Categorical	Encoded geographical area category (A–F).
VehPower	Ordered category	Encoded vehicle-power class.
VehAge	Numerical	Age of the vehicle.
DrivAge	Numerical	Age of the driver.
BonusMalus	Numerical	Bonus–malus rating associated with the policy.
VehBrand	Categorical	Encoded vehicle-brand category.
VehGas	Categorical	Vehicle fuel type: Diesel or Regular.
Density	Numerical	Population-density measure associated with the policy location.
Region	Categorical	Encoded geographical region.

Table 3.1: Variables used in the descriptive analysis

Data Quality Assessment

The dataset was checked for missing values, duplicate observations, unexpected categorical labels and unusually high numerical values before the descriptive analysis was interpreted.

Missing Values and Duplicates

No missing values were detected in any of the 12 variables. The checks also found no duplicated policy identifiers and no completely duplicated rows. The categorical variables contained the expected coded levels: six Area categories, two fuel types, 11 vehicle-brand categories and 22 regions. No blank or malformed categorical labels were identified.

These checks indicate that the dataset is complete and structurally suitable for the planned analysis. This reduces the need for missing-value treatment or duplicate removal before later statistical work.

Unusual Values and Outlier Investigation

Unusual observations were investigated rather than automatically removed. A value can be statistically extreme without being a confirmed data error, so the original data were retained unless there was clear evidence that a value was invalid.

Check	Records	Share of data	Interpretation
Exposure > 1	1,224	0.1805%	Uncommon; maximum exposure is 2.01. Retained.
ClaimNb ≥ 5	9	0.0013%	Very rare high claim counts; retained as observed claim data.
VehAge ≥ 99	48	0.0071%	Unusual ages 99/100; flagged for verification, not deleted.
DrivAge ≥ 80	8,105	1.1954%	Older-driver tail; rare but not evidence of an error by itself.
BonusMalus > 200	4	0.0006%	Extremely small high-end group; interpreted cautiously.
Density ≥ 20,000	11,308	1.6678%	High-density tail; values require contextual interpretation.

The 48 vehicle-age observations at 99 or 100 are repeated across different policies and brands, so they were flagged for later verification rather than treated as isolated typographical errors. Similarly, only four BonusMalus observations exceed 200, making any conclusions about this extreme range unstable.

Population density has a particularly long upper tail. Of the 11,308 records at or above 20,000, 10,515 have Density = 27,000. All 10,515 of these observations belong to Area F and Region R11. This systematic geographical pattern suggests that the repeated high value is associated with a specific location rather than appearing randomly across the data. The value was therefore retained.

The claim-count variable requires special care in outlier assessment. Its first quartile, median and third quartile are all zero because most observations contain no claims. For this reason, a conventional IQR rule would label ordinary positive claim counts as outliers and is not used as the main basis for removing claim observations.

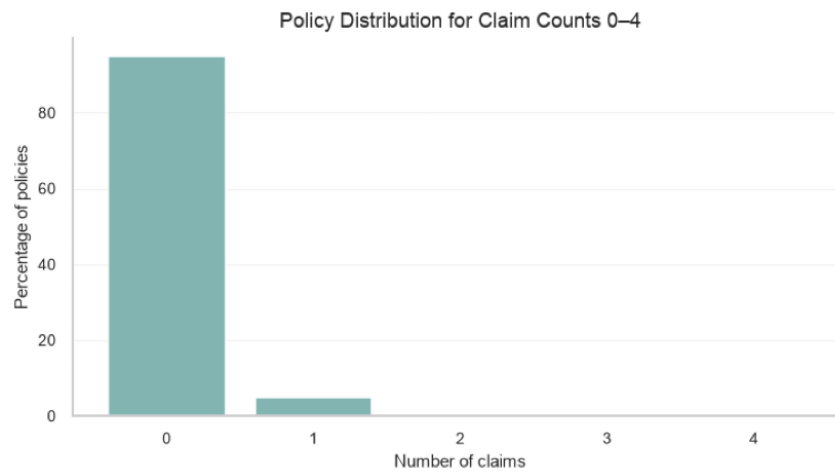
Distribution and Descriptive Analysis

Table provides the main descriptive statistics for the meaningful numerical variables. IDpol is excluded because it is an identifier.

Variable	Mean	Std. dev.	Min	Q1	Median	Q3	Max
ClaimNb	0.053	0.240	0	0	0	0	16
Exposure	0.529	0.364	0.0027	0.18	0.49	0.99	2.01
VehPower	6.455	2.051	4	5	6	7	15
VehAge	7.044	5.666	0	2	6	11	100
DrivAge	45.499	14.137	18	34	44	55	100
BonusMalus	59.762	15.637	50	50	50	64	230
Density	1,792.422	3,958.647	1	92	393	1,658	27,000

Claim Number Distribution

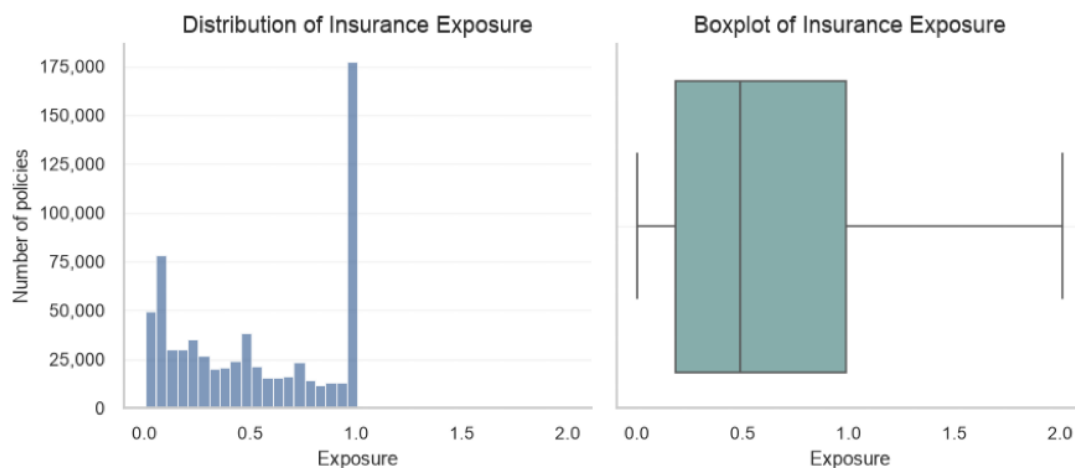
Claim counts are strongly concentrated at zero. Of the 678,013 policy observations, approximately 94.98% record no claims, about 4.75% record one claim, and multiple-claim observations become increasingly rare. Only nine observations contain five or more claims, with the maximum recorded count equal to 16.



The zero-dominated distribution is important for the later modelling stage because the response is a low-frequency count outcome rather than a continuous measurement. At this stage, the result is treated as a descriptive feature of the portfolio rather than evidence that any particular count model is automatically required.

Exposure Distribution

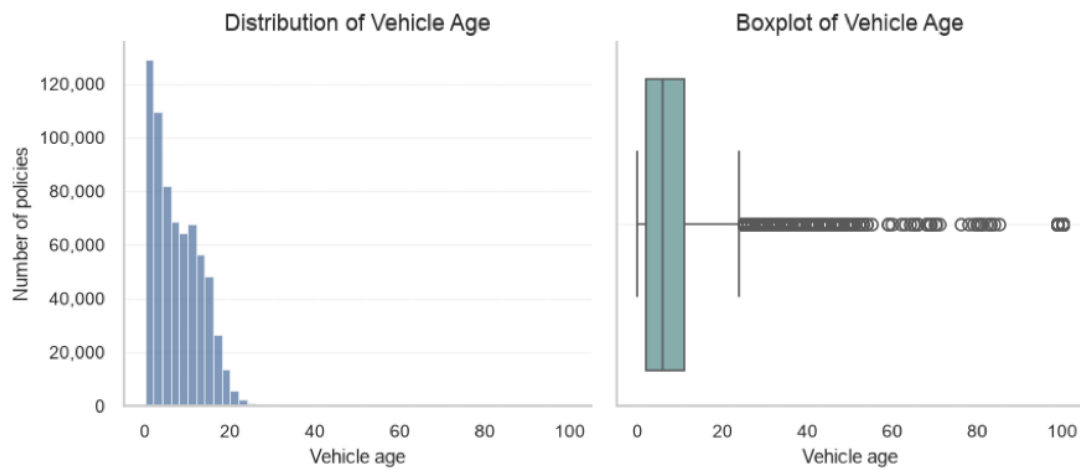
Exposure has a mean of 0.529 and a median of 0.49, with the middle 50% of observations ranging from 0.18 to 0.99. Most values are within one unit of exposure, although a small tail extends above one and reaches 2.01.



The distribution shows that policies contribute different amounts of exposure. This is why later group comparisons use claims per unit of exposure rather than raw claim totals.

Vehicle Characteristics

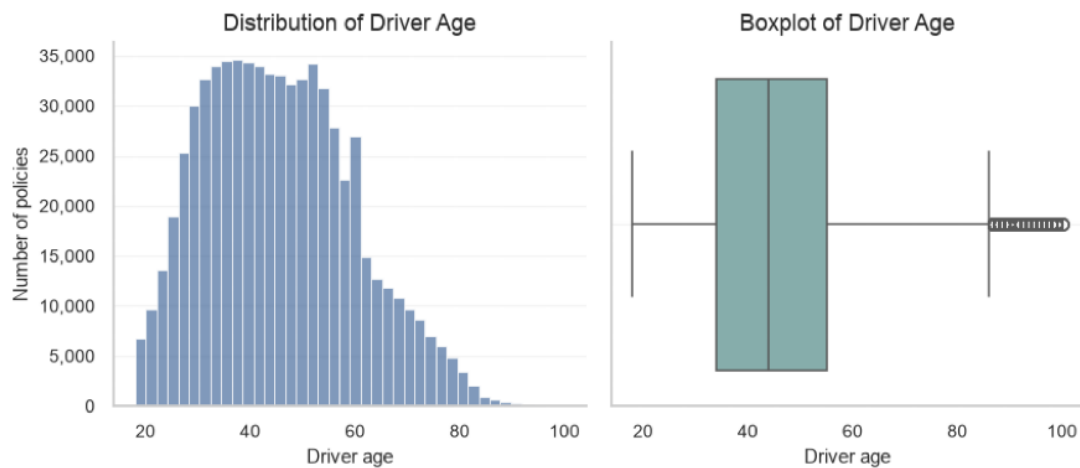
Vehicle power is concentrated mainly in classes 4–7. Classes 5, 6 and 7 contain the largest numbers of policies, while higher power classes are much less common. Vehicle age has a mean of 7.04 years and a median of 6 years. The distribution is right-skewed, with most vehicles at lower ages and a small number of much older vehicles.



The long vehicle-age tail is consistent with the unusual 99/100-year observations identified during the data-quality assessment. These extremes are visible in the boxplot but represent a very small proportion of the portfolio.

Driver Age Distribution

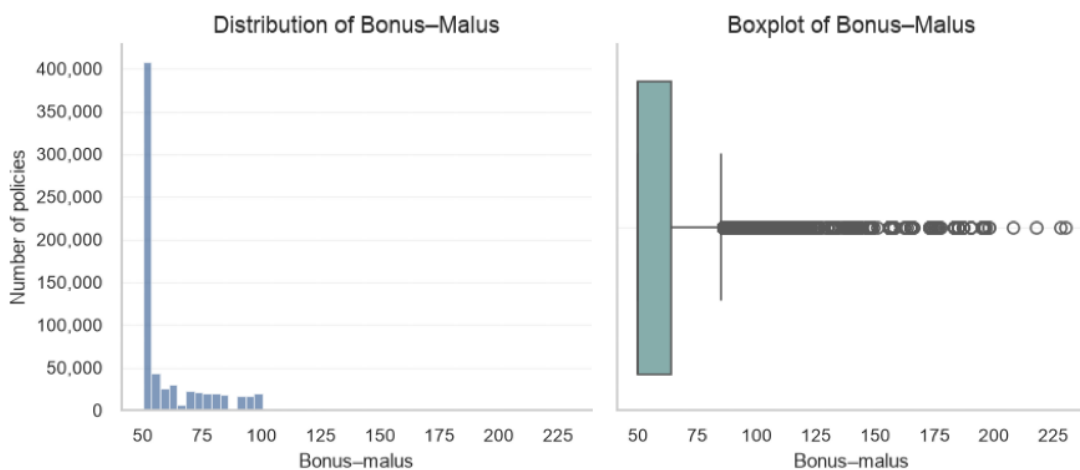
Driver age ranges from 18 to 100 years, with a mean of 45.5 and a median of 44. The largest concentration of observations is around the middle-age range, while the number of observations gradually falls at older ages.



Compared with variables such as vehicle age, BonusMalus and Density, driver age is more centrally distributed. Age groups are therefore used later to examine whether claim frequency differs across different parts of this distribution.

Bonus-Malus Distribution

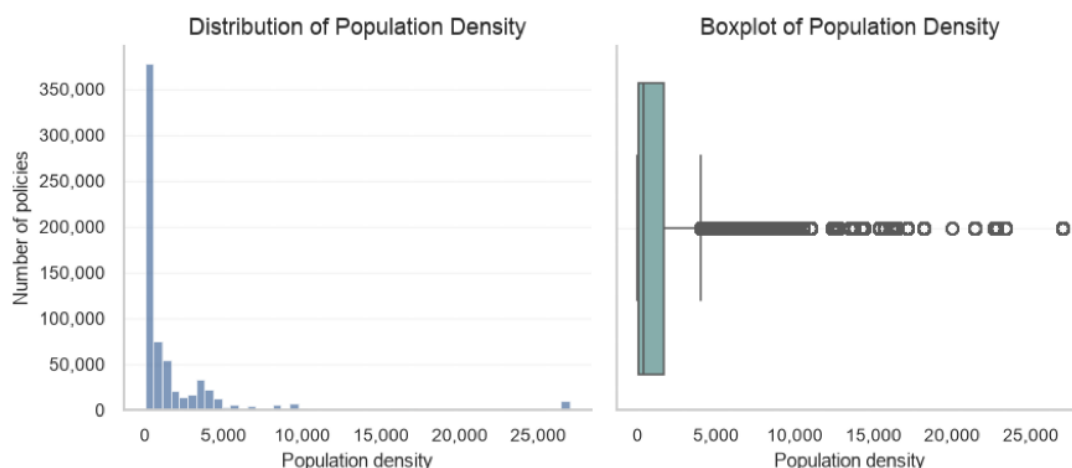
BonusMalus is strongly concentrated around its lower values. The median and first quartile are both 50, while the upper quartile is 64. The distribution has a long right tail extending to 230, but observations at the very high end are extremely rare.



This strong concentration at the lower end means that the very high BonusMalus groups contain much less data. Their claim-frequency estimates therefore need to be interpreted with greater caution than the main groups.

Population Density Distribution

Density is the most strongly right-skewed of the main explanatory variables. The median is 393, while the mean is much higher at approximately 1,792 and the maximum reaches 27,000. Most policies are associated with relatively low or moderate density values, with a smaller number of very high-density observations creating the long upper tail.



The large difference between the mean and median reflects the influence of the high-density tail. The structured Density = 27,000 group identified earlier also shows why these high values should be understood in a geographical context rather than removed solely because they appear as boxplot outliers.

Categorical Portfolio Distribution

The categorical variables show that the portfolio is not evenly distributed across all groups. Area C is the largest Area category, representing about 28.3% of observations, while Area F is the smallest at about 2.65%. Fuel type is comparatively balanced, with Regular vehicles representing about 51.0% and Diesel vehicles about 49.0%.

Vehicle brand is more concentrated: B12, B1 and B2 together account for roughly 72% of the policies. Region is also uneven, with R24 alone representing about 23.7% of the portfolio, while several regions contain less than 1%. These differences in group size are important when interpreting raw claim totals and motivate the use of exposure-adjusted claim frequency in the next section.

Descriptive Relationships with Claim Frequency

To compare groups with different portfolio sizes and exposure, observed claim frequency is calculated using total claims divided by total exposure. This measure represents claims per unit of exposure and avoids treating a group as high risk simply because it contains more policies.

$$\text{Observed Claim Frequency} = \text{Total Claims} / \text{Total Exposure}$$

The comparisons below remain descriptive. They identify patterns that may be useful for risk segmentation and later modelling, but they do not establish causation or statistical significance.

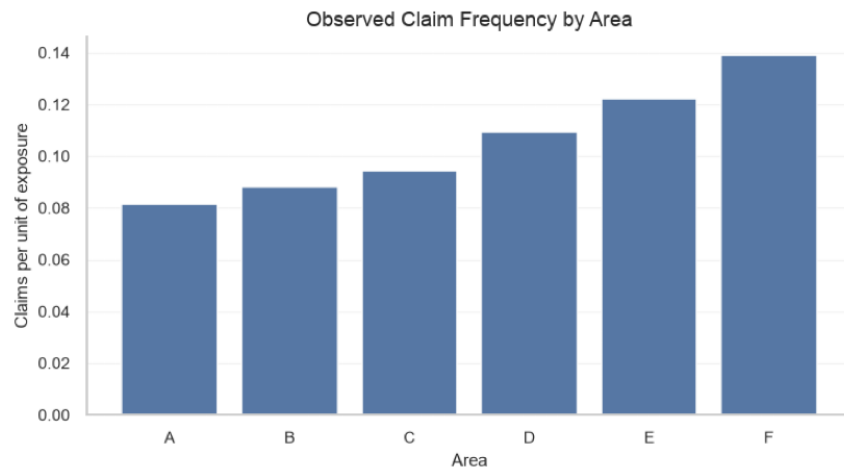
Claim Frequency by Area

Area shows a clear ordered pattern. Observed claim frequency increases from 0.0817 in Area A to 0.0883 in B, 0.0945 in C, 0.1093 in D, 0.1223 in E and 0.1391 in F.

This pattern indicates that geographical area may contain useful information for portfolio segmentation and risk monitoring. However, Area is an encoded geographical measure and may overlap with other geographical information such as Density and Region, so its independent contribution needs to be assessed later.

Claim Frequency by Driver Age

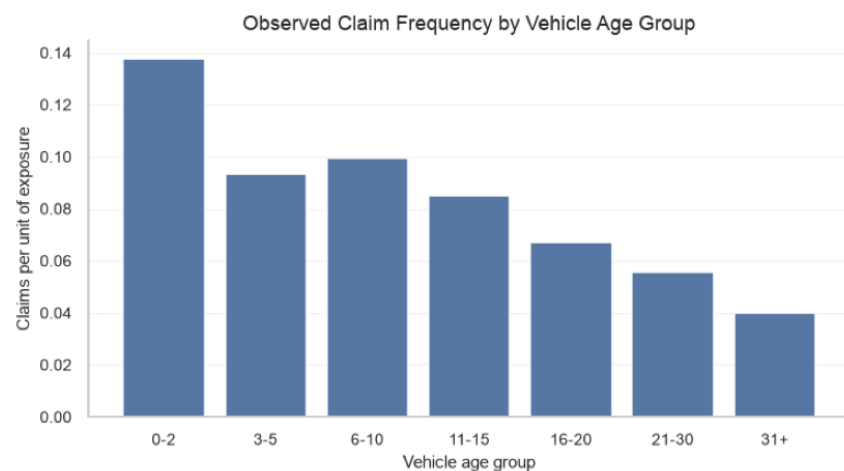
Driver age shows one of the clearest differences in the descriptive analysis. The 18–24 group has an observed claim frequency of 0.1893. All other age groups are much lower, ranging from approximately 0.0912 to 0.1059, with the 75+ group at 0.0997.



The youngest group therefore stands out as an important segment for later statistical investigation. The result could be relevant for underwriting and portfolio monitoring, but it should not be interpreted as evidence that driver age alone causes the difference in claim frequency.

Claim Frequency by Vehicle Age

Vehicle age shows an overall declining pattern in observed claim frequency. Vehicles aged 0–2 years have the highest frequency at 0.1377. The frequency is 0.0936 for ages 3–5, 0.0994 for 6–10, 0.0852 for 11–15, 0.0670 for 16–20, 0.0558 for 21–30 and 0.0400 for the 31+ group.



The pattern is not perfectly monotonic between every adjacent category, but the overall direction suggests that vehicle age may be useful in later risk modelling. The 31+ group contains only 1,116 policies and much less exposure than the main age groups, so its low estimate should be interpreted cautiously.

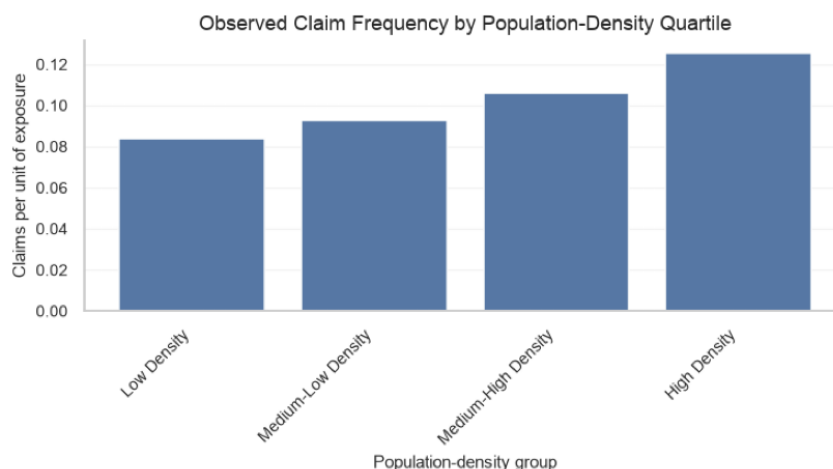
Claim Frequency by Bonus-Malus

BonusMalus shows a strong positive descriptive relationship with claim frequency across the main groups. Observed frequency increases from 0.0824 for 50–59 to 0.1266 for 60–69, 0.1321 for 70–79, 0.1430 for 80–99 and 0.3097 for 100–149.

The two highest groups produce frequencies of 0.5512 for 150–200 and 2.5316 for 201+, but these estimates are based on only 205 and four policies respectively. The 201+ group also has only 0.79 units of total exposure. These extreme estimates are therefore not treated as stable evidence about the wider portfolio. The main-group pattern nevertheless indicates that BonusMalus is likely to be an important variable for further statistical investigation and risk segmentation.

Claim Frequency by Population Density

When Density is divided into four approximately equal-sized groups, observed claim frequency increases consistently with density. The frequencies are 0.0841 for the low-density group, 0.0929 for medium-low, 0.1059 for medium-high and 0.1252 for the high-density group.



The result supports the broader geographical pattern observed for Area. More densely populated locations are associated with higher observed claim frequency in this dataset. Because Area, Region and Density may describe related geographical characteristics, later modelling should determine how much independent information each variable contributes.

Other Group Comparisons

The remaining group comparisons show useful but less straightforward patterns. Fuel type has only a small descriptive difference: Diesel has an observed frequency of 0.0976 and Regular 0.1036. Vehicle brand varies more widely, from 0.0730 for B14 to 0.1367 for B12. Region also varies, from 0.0753 in R41 to 0.1399 in R94, although several of the more extreme regional values are based on smaller amounts of exposure.

Vehicle power does not show a simple increasing or decreasing relationship. Frequencies range from 0.0847 for class 8 to 0.1164 for class 10, with irregular movement across the remaining classes. This indicates that vehicle power should not be reduced to a simple assumption that higher power automatically corresponds to higher claim frequency.

Initial Business Insights

The descriptive analysis provides several initial findings that are relevant to the organization. First, the portfolio is dominated by zero-claim observations, with approximately 95% of policies recording no claims. This confirms that claim frequency is a low-frequency count problem and will be important when appropriate statistical models are evaluated later.

Geographical characteristics appear to be important. Area shows a progressive increase in observed frequency from A to F, Density shows a similar increase across quartile groups, and Region also displays noticeable variation. These findings suggest that location-related information may be useful for risk segmentation and portfolio monitoring, although the geographical variables may contain overlapping information.

Driver age also produces a clear descriptive signal, with the 18–24 group showing substantially higher observed claim frequency than the older groups. Vehicle age shows an overall opposite pattern, with newer vehicles recording higher observed frequency and older groups generally recording lower frequency. Both variables therefore warrant further statistical investigation.

BonusMalus provides another strong relationship across the main groups, with claim frequency increasing as the rating rises. The very highest BonusMalus categories are too small to support strong conclusions, but the main pattern suggests that this variable may be particularly useful for later modelling and risk classification.

Not all variables show equally strong patterns. Fuel type produces only a small difference, while vehicle power changes irregularly across categories. This is useful because it prevents the organization from assuming that every available variable contributes equally to risk segmentation.

These findings are exploratory. They identify areas that may support underwriting, portfolio monitoring and later risk modelling, but they do not yet justify firm management or pricing decisions. Statistical inference is required to determine whether the observed differences are statistically supported, and predictive modelling is required to assess their combined contribution.

Summary

The freMTPL2freq dataset is complete with respect to missing values and duplicate records, although several rare or extreme numerical observations require careful interpretation. The distributions show a strongly zero-dominated claim-count outcome, right-skewed vehicle age, BonusMalus and population density, and uneven representation across several categorical groups.

Exposure-adjusted comparisons indicate the strongest initial descriptive patterns for Area, driver age, vehicle age, BonusMalus and population density. Vehicle brand and Region also show variation, while fuel type is comparatively similar and vehicle power has no simple ordered pattern. These results provide the descriptive foundation for Task 4, where appropriate statistical inference techniques can be used to assess whether the observed differences are statistically supported.

Task 4 – Statistical Inference

Introduction

Task 3 identified several differences in claim behaviour across policyholder, vehicle and geographical characteristics. However, descriptive results alone cannot determine whether these differences are statistically supported or may be due to random variation.

Task 4 therefore applies suitable statistical inference techniques to the main patterns identified in Task 3. For each analysis, the hypothesis, test selection, statistical result and practical meaning for the insurer are considered. Statistical significance is interpreted together with the size of the observed difference because the dataset contains 678,013 policy observations and very small effects can become statistically significant in a sample of this size.

Basis for Selecting the Analyses

The response variable ClaimNb is a strongly zero-dominated count variable, with approximately 94.98% of policies recording no claims. This means that not every standard inference method is suitable for the main claim-frequency outcome. Proportion tests are used when claim experience is simplified to claim/no claim, a Welch t-test is used for a comparison of mean BonusMalus, and Poisson rate models are used when the actual claim count and insurance Exposure must be considered.

Observed Claim Frequency = Total Claims / Total Exposure

The selected questions are based directly on the patterns identified in Task 3. Area, driver age, vehicle age, BonusMalus and Density showed clear descriptive differences, while fuel type showed a smaller difference and is useful for comparing statistical significance with practical importance.

Analysis	Statistical question	Method
1	Does claim occurrence differ between Diesel and Regular vehicles?	Two-sample proportion test
2	Is claim occurrence associated with Area?	Chi-square test of independence
3	Does average BonusMalus differ between claim and no-claim policies?	Welch two-sample t-test
4	Does claim rate differ across driver-age groups?	Poisson rate model with Exposure
5	Does claim rate differ across vehicle-age groups?	Poisson rate model with Exposure
6	Are Area, BonusMalus and Density related to exposure-adjusted claim rate?	Poisson rate models with Exposure

Consideration of ANOVA and Variance Comparisons

A separate comparison of variances and ordinary ANOVA were considered but were not applied to ClaimNb. The response is a strongly zero-dominated count variable rather than a continuous approximately normal measurement, so standard variance tests and ordinary ANOVA would not be well suited to the main claim-frequency outcome. Variation in the count response is instead considered through count-model dispersion diagnostics, including an assessment of overdispersion during Task 5.

Analysis 1 – Comparison of Claim Occurrence by Vehicle Fuel Type

Fuel type was selected because Task 3 showed only a small descriptive difference between Diesel and Regular vehicles. The analysis tests whether the proportion of policies experiencing at least one claim differs between the two fuel types.

Hypotheses and Method

H0: $p_{\text{Diesel}} = p_{\text{Regular}}$

H1: $p_{\text{Diesel}} \neq p_{\text{Regular}}$

A two-sample proportion test was used because the analysis compares two groups using a binary outcome: no claim or at least one claim.

Fuel type	Policies	Policies with claim	Claim proportion
Diesel	332,136	15,709	4.73%
Regular	345,877	18,351	5.31%

Statistical Result and Interpretation

The test produced $\chi^2 = 117.68$ with 1 degree of freedom and $p < 2.2 \times 10^{-16}$. The null hypothesis was therefore rejected at the 5% significance level. The 95% confidence interval for the Diesel minus Regular difference was -0.0068 to -0.0047, which also indicates a lower claim-occurrence proportion for Diesel vehicles.

Regular vehicles had a claim occurrence proportion of 5.31% compared with 4.73% for Diesel vehicles, a difference of approximately 0.58 percentage points. Although statistically significant, the difference is small in practical terms. Fuel type may therefore provide some information about claim behaviour but should not be used alone for risk segmentation.

Analysis 2 – Association Between Claim Occurrence and Area

Area was selected because Task 3 showed a progressive increase in observed claim frequency from Area A to Area F. The analysis tests whether claim occurrence is statistically associated with the six Area categories.

Hypotheses and Method

H0: Claim occurrence is independent of Area

H1: Claim occurrence is associated with Area

A Pearson chi-square test of independence was used because both variables are categorical: Area has six categories and claim occurrence has two outcomes.

Area	Policies	Policies with claim	Claim proportion
A	103,957	4,831	4.65%
B	75,459	3,630	4.81%
C	191,880	9,356	4.88%
D	151,596	7,892	5.21%
E	137,167	7,296	5.32%
F	17,954	1,055	5.88%

Statistical Result and Interpretation

The test produced $\chi^2 = 109.84$ with 5 degrees of freedom and $p < 2.2 \times 10^{-16}$. The null hypothesis was rejected, providing strong statistical evidence that claim occurrence is associated with Area.

Claim occurrence generally increased from 4.65% in Area A to 5.88% in Area F. This supports the geographical pattern identified in Task 3, although the result does not establish that Area itself causes claims. Area may also represent related conditions such as population density, traffic environment or other regional characteristics.

Analysis 3 – Comparison of Mean BonusMalus Between Claim and No-Claim Policies

BonusMalus was selected because Task 3 showed a strong positive descriptive relationship between BonusMalus groups and claim frequency. This analysis tests whether the average BonusMalus differs between policies with and without at least one claim.

Hypotheses and Method

H0: $\mu_{\text{No claim}} = \mu_{\text{Claim}}$

H1: $\mu_{\text{No claim}} \neq \mu_{\text{Claim}}$

A Welch two-sample t-test was used because two independent groups are compared on the numerical BonusMalus variable. Welch's version does not require the two groups to have equal variances.

Group	Policies	Mean BonusMalus	SD
No claim	643,953	59.59	15.4
At least one claim	34,060	63.10	19.0

Statistical Result and Interpretation

The Welch t-test produced $t = -33.591$ with approximately 36,479 degrees of freedom and $p < 2.2 \times 10^{-16}$. The 95% confidence interval for the no-claim minus claim mean difference was -3.72 to -3.31. The null hypothesis was therefore rejected.

Policies with at least one claim had an average BonusMalus of 63.10 compared with 59.59 for no-claim policies, a difference of approximately 3.51 points. This result supports BonusMalus as a potentially useful risk indicator, although the variable is right-skewed and the analysis remains a single-variable comparison rather than evidence of causation.

Analysis 4 – Driver Age and Exposure-Adjusted Claim Rate

Driver age was selected because the 18–24 group showed one of the clearest descriptive differences in Task 3. A Poisson rate model was used so that the actual number of claims could be analysed while differences in insurance Exposure were taken into account.

Hypotheses and Method

H0: Claim rate does not differ across driver-age groups

H1: At least one driver-age group has a different claim rate

The model used ClaimNb as the count response and log(Exposure) as an offset. The 18–24 group was used as the reference category. A likelihood-ratio analysis of deviance compared the age-group model with a model containing Exposure only.

Driver age	Policies	Claim frequency	Rate ratio vs 18–24
18–24	30,198	0.1893	1.000
25–34	140,947	0.0990	0.523
35–44	170,578	0.0932	0.492
45–54	164,034	0.1059	0.559
55–64	99,094	0.0912	0.482
65–74	50,821	0.0927	0.490
75+	22,341	0.0997	0.527

Statistical Result and Interpretation

Adding driver-age group reduced residual deviance from 224,481 to 223,562. The likelihood-ratio test gave a deviance difference of 919.55 with 6 degrees of freedom and $p < 2.2 \times 10^{-16}$. Driver-age group is therefore statistically associated with claim rate after accounting for Exposure.

The 18–24 group had an observed rate of 0.1893 and was substantially higher than every older group. Relative to 18–24, the older groups had rate ratios between approximately 0.482 and 0.559, corresponding to claim rates roughly 44%–52% lower. This identifies the youngest driver group as an important segment for further multivariable modelling and portfolio monitoring.

Analysis 5 – Vehicle Age and Exposure-Adjusted Claim Rate

Vehicle age was selected because Task 3 showed an overall decline in observed claim frequency as vehicle age increased. The same Poisson rate approach was used to test whether this pattern is statistically supported after accounting for Exposure.

Hypotheses and Method

H0: Claim rate does not differ across vehicle-age groups

H1: At least one vehicle-age group has a different claim rate

The 0–2 year group was used as the reference category. ClaimNb was modelled as a Poisson count with log(Exposure) included as an offset.

Vehicle age	Policies	Claim frequency	Rate ratio vs 0–2
0–2	188,147	0.1377	1.000
3–5	132,490	0.0936	0.680
6–10	171,594	0.0994	0.722
11–15	134,063	0.0852	0.619
16–20	43,397	0.0670	0.487
21–30	7,206	0.0558	0.406
31+	1,116	0.0400	0.291

Statistical Result and Interpretation

Adding vehicle-age group reduced residual deviance from 224,481 to 222,821. The likelihood-ratio test gave a deviance difference of 1,660.2 with 6 degrees of freedom and $p < 2.2 \times 10^{-16}$. Vehicle age is therefore statistically associated with claim rate after accounting for Exposure.

Compared with vehicles aged 0–2 years, the estimated rate ratio was 0.680 for ages 3–5 and declined to 0.291 for the 31+ group. This means the 31+ group had an estimated claim rate about 71% lower than the 0–2 group. The pattern is not perfectly monotonic between every adjacent category, but the overall relationship supports vehicle age as a useful candidate for later modelling. The oldest group is small and should be interpreted cautiously.

Analysis 6 – Exposure-Adjusted Rate Comparisons for Area, BonusMalus and Density

The final inferential analysis applies the same exposure-adjusted Poisson approach to three variables that showed clear descriptive relationships in Task 3. Each variable is tested separately to determine whether the observed claim-rate pattern is statistically supported before multivariable modelling.

6.1 Area

H0: Claim rate does not differ across Areas

H1: At least one Area has a different claim rate

A Poisson rate model with $\log(\text{Exposure})$ as an offset was used because ClaimNb is a count response and policy observations contribute different amounts of insurance exposure.

Area A was used as the reference category. Adding Area reduced residual deviance from 224,481 to 223,698. The likelihood-ratio test produced a deviance difference of 783.55 with 5 degrees of freedom and $p < 2.2 \times 10^{-16}$. The null hypothesis was therefore rejected, providing strong evidence that exposure-adjusted claim rates differ across Areas.

Area	Observed claim frequency	Rate ratio vs A
A	0.0817	1.000
B	0.0883	1.081
C	0.0945	1.157
D	0.1093	1.338
E	0.1223	1.497
F	0.1391	1.703

The rate increased steadily from Area A to F. Relative to Area A, the estimated rate ratios were 1.081 for B, 1.157 for C, 1.338 for D, 1.497 for E and 1.703 for F. Area F therefore had an estimated claim rate about 70.3% higher than Area A. This strengthens the earlier evidence that geography is relevant to claim frequency, but overlap with Density and Region must be considered in later multivariable analysis.

6.2 BonusMalus

H0: Claim rate does not differ across BonusMalus groups

H1: At least one BonusMalus group has a different claim rate

A Poisson rate model with $\log(\text{Exposure})$ as an offset was used to compare claim rates across the BonusMalus groups while accounting for different amounts of exposure.

The 50–59 BonusMalus group was used as the reference category. Adding BonusMalus group reduced residual deviance from 224,481 to 219,892. The likelihood-ratio test produced a deviance difference of 4,589.5 with 6 degrees of freedom and $p < 2.2 \times 10^{-16}$. The null hypothesis was therefore rejected, confirming that exposure-adjusted claim rates differ across BonusMalus groups.

BonusMalus group	Observed claim frequency	Rate ratio vs 50–59
50–59	0.0824	1.000
60–69	0.1266	1.536
70–79	0.1321	1.602
80–99	0.1430	1.735
100–149	0.3097	3.756
150–200	0.5512	6.685
201+	2.5316	30.706

The main groups show a strong increasing relationship between BonusMalus and claim rate. Relative to the 50–59 group, the estimated rate ratios were 1.536 for 60–69, 1.602 for 70–79, 1.735 for 80–99 and 3.756 for 100–149. The 150–200 and 201+ groups produced much larger ratios of 6.686 and 30.706, but these categories contain only 205 and four policies respectively, so the extreme estimates are unstable and should not be generalized to the wider portfolio.

6.3 Population Density

H0: Claim rate does not differ across population-density groups

H1: At least one population-density group has a different claim rate

A Poisson rate model with $\log(\text{Exposure})$ as an offset was used to test the Density pattern identified in Task 3 while accounting for insurance exposure.

Density was divided into four approximately equal-sized groups, with Low Density used as the reference. The likelihood-ratio test gave a deviance difference of 802.75 with 3 degrees of freedom and $p < 2.2 \times 10^{-16}$.

Density group	Observed claim frequency	Rate ratio vs Low
Low	0.0841	1.000
Medium-Low	0.0929	1.104
Medium-High	0.1059	1.258
High	0.1252	1.488

Claim rate increased consistently with population density. The High Density group had an estimated claim rate approximately 48.8% higher than the Low Density group. This provides statistical support for the density pattern observed in Task 3 and further suggests that geographical conditions may be important for portfolio risk segmentation.

Limitations of the Inferential Analysis

The analyses in this task are mainly single-variable comparisons. Statistical significance therefore does not imply that a variable has an independent effect after other characteristics are considered, and none of the results should be interpreted as causal. The very large dataset also means that small effects can produce extremely small p-values, so practical magnitude was considered alongside significance.

The Poisson rate analyses assume a Poisson count structure. These results are treated as initial inferential evidence because substantial overdispersion could affect standard errors and significance tests. Overdispersion and the suitability of alternatives such as Negative Binomial regression will therefore be assessed before the count models are used for final predictive conclusions in Task 5. The simple proportion and chi-square analyses also do not adjust for Exposure, which is why exposure-adjusted Poisson models were used for the main claim-rate comparisons.

Summary

The statistical inference results provide support for several patterns identified in Task 3. Fuel type showed a statistically significant but small difference in claim occurrence. Area was significantly associated with claim occurrence, while policies with claims had a significantly higher average BonusMalus than no-claim policies.

The exposure-adjusted Poisson analyses showed statistically significant differences in claim rate across driver-age groups, vehicle-age groups, Area, BonusMalus groups and population-density groups. The clearest practical patterns included the high claim rate among drivers aged 18–24, higher rates for newer vehicles, increasing rates across Area and Density categories, and a strong increase across the main BonusMalus groups.

These findings provide an inferential foundation for Task 5. The next stage should determine whether these variables remain useful when considered together, evaluate count-model assumptions such as overdispersion, and compare suitable predictive models before any final organizational recommendation is made.

Task 5 – Predictive Statistical Modelling

Objective. To predict motor-insurance claim frequency using policy, driver, vehicle and geographical characteristics while accounting for differences in policy exposure. The response variable is ClaimNb and Exposure is included as an offset.

5.1 Modelling Strategy

The analysis used an 80/20 train-test split. The training set contained 542,410 observations and the remaining 135,603 observations were reserved for out-of-sample evaluation. The split was stratified by whether a policy recorded at least one claim so that the claim/no-claim balance remained similar in both samples.

Five modelling approaches were considered: an exposure-only Poisson baseline, a multivariable Poisson GLM, Negative Binomial regression, LASSO Poisson regression and Elastic Net Poisson regression. The baseline provides a simple benchmark, while the regularized models assess whether a smaller or more strongly shrunk coefficient set can retain predictive performance.

Predictors included Area, VehPower, VehAge, DrivAge, BonusMalus, VehBrand, VehGas, log-transformed population density and Region. IDpol was excluded because it is an identifier. Density was transformed using $\log(1 + \text{Density})$ because of its strong right-skew.

5.2 Baseline and Multivariable Poisson Model

The exposure-only baseline assumed a common underlying claim rate for all policies and used only $\log(\text{Exposure})$ as an offset. Its AIC was 235,138.39. The full Poisson model reduced AIC to 229,270.00 and reduced test-set Poisson deviance from 0.330343 to 0.319797, an improvement of approximately 3.2% over the baseline.

The multivariable model therefore provided useful predictive information beyond exposure alone. The test-set predicted claim frequency was 0.100758 compared with an observed frequency of 0.101044, a difference of only about 0.28%.

Term	Rate ratio	p-value	Interpretation
VehAge	0.9615	<0.001	Each additional vehicle year is associated with about 3.9% lower expected claim frequency.
DrivAge	1.0065	<0.001	A one-year increase is associated with about 0.65% higher expected

			frequency under the linear specification.
BonusMalus	1.0228	<0.001	Each one-point increase is associated with about 2.3% higher expected frequency; a 10-point increase is about 25% higher.
VehGas: Regular	1.0521	<0.001	Regular-fuel vehicles have about 5.2% higher expected frequency than Diesel, controlling for the other variables.
VehBrand: B12	1.1941	<0.001	B12 has about 19.4% higher expected frequency than the reference brand.
VehPower: 9	1.3154	<0.001	Power class 9 has about 31.5% higher expected frequency than the reference power class.
LogDensity	1.0315	0.030	Higher population density remains positively associated with claim frequency after adjustment.

Table 5.1: Selected effects from the multivariable Poisson GLM

An important multivariable result is that the Area coefficients were no longer statistically significant after Region and Density were included. This differs from the earlier unadjusted comparisons and suggests that some of the Area-level pattern is shared with more detailed geographical information. This is a useful example of why the predictors need to be considered jointly rather than one at a time.

The positive linear DrivAge coefficient should be interpreted cautiously. Earlier descriptive analysis showed the youngest drivers as a distinct high-risk group, so a single straight-line age effect may be too simple. This is a limitation of the current specification and supports testing a nonlinear age effect in a later refinement.

5.3 Poisson Diagnostics

The Pearson dispersion statistic was 2.676. A value near 1 would be expected under a well-fitting Poisson variance assumption, so this result provides evidence of overdispersion. This means the standard Poisson model understates the variability in the count outcome, even though its mean predictions are well calibrated.

The zero-count check did not indicate a large excess-zero problem. The observed proportion of zero claims in the training data was 94.976%, while the Poisson model implied 94.910%. The difference is only about 0.07 percentage points. Therefore, the large number of zeros is largely consistent with the low claim rate and does not, by itself, justify a zero-inflated model.

5.4 Negative Binomial Alternative

Negative Binomial regression was fitted because the Poisson model showed overdispersion. However, the current Negative Binomial estimation did not converge: the model returned an infinite dispersion parameter, NaN log-likelihood and NaN standard errors. These results are not statistically valid and should not be used for model selection.

The identical test predictions shown for the failed Negative Binomial fit and the Poisson model should therefore be treated as an estimation artifact rather than evidence that the two models perform equally. A converged Negative Binomial model should be obtained before the final submission, for example by

rescaling predictors, simplifying the geographical structure, using alternative starting values/optimizers, or fitting a GLM Negative Binomial specification with a stable dispersion estimate.

5.5 Regularized Poisson Models

LASSO and Elastic Net Poisson models were used as regularized alternatives. Continuous predictors were standardized before fitting. Validation selected $\alpha = 0.0001$ for both approaches. The LASSO solution retained a smaller set of non-zero coefficients, with BonusMalus, VehAge, DrivAge, LogDensity and selected vehicle/geographical categories remaining in the model.

The regularized models did not improve test Poisson deviance over the full Poisson GLM. This suggests that the additional shrinkage did not provide a predictive advantage for this dataset at the selected penalty level, although LASSO remains useful for identifying a more compact coefficient set.

5.6 Out-of-Sample Model Comparison

Model	MAE	RMSE	Poisson deviance	Predicted frequency	AIC (training)
Exposure-only baseline	0.099955	0.240951	0.330343	0.100618	235,138.39
Poisson GLM	0.098834	0.239628	0.319797	0.100758	229,270.00
Negative Binomial	Invalid	Invalid	Invalid	Invalid	Not available – model did not converge
LASSO Poisson	0.099011	0.239652	0.320328	0.100955	Not directly compared
Elastic Net Poisson	0.099015	0.239613	0.320396	0.101098	Not directly compared

Table 5.2: Predictive model comparison on the test set

Among the successfully estimated models, the standard multivariable Poisson GLM produced the lowest test Poisson deviance and MAE. Elastic Net produced a marginally lower RMSE, but the difference was extremely small and it performed slightly worse on Poisson deviance. Both regularized models remained close to the full Poisson model, indicating that their shrinkage did not produce a clear out-of-sample gain.

5.7 Calibration

Calibration was assessed by dividing the test policies into five groups according to predicted claim risk. Observed and predicted claim frequencies were close across the complete risk range, including the highest-risk group.

Risk group	Observed frequency	Predicted frequency
1	0.055812	0.057765
2	0.077438	0.078174
3	0.097508	0.095294
4	0.118810	0.117341
5	0.193316	0.191887

Table 5.3: Poisson model calibration by predicted risk group



Figure 5.1: Observed and predicted claim frequency by Poisson risk group

5.8 Strengths, Limitations and Model Recommendation

Strengths. The modelling framework directly matches the count nature of ClaimNb, correctly adjusts for different exposure periods, evaluates performance on unseen data, and provides interpretable rate ratios. The calibration results also show that the model separates low- and high-risk policy groups while maintaining close agreement between observed and predicted frequencies.

Limitations. The Poisson variance assumption is not satisfied because dispersion is substantially above 1. Several continuous effects are currently represented linearly even though variables such as driver age may be nonlinear. Geographical variables may contain overlapping information. In addition, the Negative Binomial model has not yet produced a valid converged fit, so the comparison between Poisson and a true overdispersion model is incomplete.

Current recommendation. Among the models that fitted successfully, the multivariable Poisson GLM currently provides the strongest overall balance of predictive performance, calibration and interpretability. It improves clearly on the exposure-only baseline and slightly outperforms the regularized alternatives on Poisson deviance. However, because the dispersion statistic is 2.676, this recommendation should be treated as provisional until a stable Negative Binomial fit is obtained. If a converged Negative Binomial model does not materially improve test performance or calibration, the Poisson structure can remain the operational frequency model, with overdispersion handled appropriately for statistical inference.

Organizational use. The selected model can support risk segmentation, underwriting review and portfolio monitoring by estimating expected claim frequency from available policy characteristics. It should not be interpreted as a complete premium-setting model because claim severity, operating costs, profit margins and other commercial pricing components are outside the scope of the analysis.

Recommended final refinement before submission: obtain a converged Negative Binomial model and test a nonlinear representation of driver age (for example age bands or a spline/GAM term). These two checks directly address the main diagnostic weaknesses identified in the current results.

Task 6 – Critical Evaluation of Experimental Design

Introduction

The current freMTPL2freq dataset is historical and observational, so the analysis in earlier tasks can identify associations but cannot directly establish whether an intervention causes a change in claim behaviour. Experimental design could be useful in a future investigation where the insurer actively introduces a risk-reduction programme and randomly assigns eligible drivers to treatment conditions.

For this task, a future safe-driving digital intervention is considered. The proposed programme would use telematics-based driving information to provide regular safety feedback and reminders to selected policyholders, while a control group would continue to receive the insurer's normal service. The main business question is whether the intervention can reduce future claim frequency or improve driving behaviour.

Proposed Future Experiment

Component	Proposed approach
Experimental unit	Eligible insured driver / policyholder
Treatment	Telematics-based safe-driving feedback and reminders
Control	Normal insurance service without the added intervention
Primary outcome	Future claim frequency per unit of exposure
Secondary outcomes	Telematics measures such as speeding, harsh braking or rapid acceleration

Randomized field studies provide evidence that telematics feedback can be tested experimentally. Stevenson et al. (2021) randomly allocated drivers to feedback, feedback plus incentives, and control groups, while more recent randomized studies have also reported improvements in driving-safety measures from telematics-based

feedback and incentive programmes (Malekpour et al., 2023; Ebert et al., 2026). These studies support the practical idea of evaluating a digital driving intervention using randomized experimental methods.

Completely Randomized Design (CRD)

In a Completely Randomized Design, treatments are assigned to experimental units entirely at random. This design is most appropriate when the experimental units are reasonably comparable and there is no major nuisance factor that needs to be controlled before randomization.

Application to the Insurance Problem

Under a CRD, eligible drivers would be randomly assigned to either the safe-driving intervention or the control group. Randomization would give each eligible driver a comparable chance of receiving the intervention and would help distribute measured and unmeasured driver characteristics across the two groups.

Advantages

The main advantage of CRD is simplicity. It is straightforward to implement, explain and monitor, and random assignment reduces systematic selection bias between the treatment and control groups. If the groups are sufficiently large, factors such as age, vehicle type and location should be reasonably balanced through randomization.

Limitations and Implementation Challenges

The main limitation is the heterogeneity of the insurance portfolio. Earlier analysis showed substantial differences in claim frequency across driver age, vehicle age, Area, BonusMalus and Density. Pure randomization may still produce some imbalance in important risk characteristics, particularly in a smaller pilot. Claim events are also relatively rare, so a large sample and sufficient follow-up would be needed if claim frequency is used as the main outcome. Customer consent, privacy of telematics data, intervention adherence and implementation cost would also need to be managed.

Randomized Complete Block Design (RCBD)

A Randomized Complete Block Design first divides experimental units into blocks based on an important source of variation and then randomizes the treatments within each block. Every block contains all treatment conditions. Blocking is intended to reduce unwanted variation and improve the precision of the treatment comparison.

Application to the Insurance Problem

Driver age is a strong candidate for the blocking factor because the earlier analysis found clear differences in claim frequency between age groups, especially for drivers aged 18-24. Drivers could first be separated into age blocks and then randomly assigned within each block to either the intervention or control condition. This would ensure that both treatment conditions contain drivers from each age category.

Example age block	Randomization within block
18-24	Intervention or Control
25-34	Intervention or Control
35-44	Intervention or Control
45-54	Intervention or Control
55+	Intervention or Control

Advantages

RCBD would directly control one known source of risk variation rather than relying on complete randomization to balance it. This could provide a more precise estimate of the intervention effect when driver age strongly influences claim frequency. Other variables such as Area or BonusMalus could also be considered as blocking factors in alternative designs if they are more relevant to the specific intervention.

Limitations and Implementation Challenges

RCBD is more complex to organize than CRD. The insurer must choose a suitable blocking variable before treatment assignment, maintain adequate numbers of participants in each block, and ensure that both treatment conditions are represented within every block. Too many or very small blocks can make implementation difficult. A single-blocking-factor RCBD also cannot control every source of portfolio variation at the design stage.

Comparison of CRD and RCBD

Aspect	CRD	RCBD
Randomization	Across all eligible drivers	Within pre-defined blocks
Complexity	Lower	Higher
Control of known risk differences	Relies mainly on randomization	Directly controls the chosen blocking factor
Suitability for this portfolio	Suitable for a large, simple pilot	Useful when a strong risk factor such as age should be balanced
Main weakness	Possible imbalance in heterogeneous drivers	Requires correct block selection and enough drivers per block

Practical Feasibility

A digital safe-driving intervention is practically feasible as a pilot because telematics and smartphone-based feedback have already been evaluated in randomized driving studies. However, claim frequency is a low-frequency outcome, so an experiment focused only on insurance claims may require a large sample and substantial follow-up. Telematics measures such as speeding, harsh braking and rapid acceleration could therefore be used as secondary outcomes to provide earlier evidence of behavioural change while claim frequency remains the main insurance outcome.

The experiment would also require clear consent procedures, secure handling of driving data, consistent delivery of the intervention and monitoring of participant dropout. These operational and ethical requirements could increase cost, but a controlled pilot would provide stronger causal evidence than relying only on historical observational data.

Recommended Design

For this insurance problem, RCBD would generally be preferable if the insurer can identify one major risk characteristic to control before randomization. Driver age is a reasonable blocking factor because it showed a strong relationship with claim frequency in the earlier analysis. Randomizing treatment and control drivers within age blocks would reduce the chance that the estimated intervention effect is distorted by differences in the age composition of the groups.

CRD would still be suitable for a large and operationally simple trial where complete randomization is expected to produce well-balanced groups. The final choice should therefore depend on the size of the pilot, the available customer population, the strength of known risk differences and the insurer's ability to manage blocking in practice.

Statistical Analysis of a Future Experiment

The experimental design does not require the final outcome to be analysed using ordinary ANOVA. Because insurance claims are counts and exposure differs between policies, a future experiment could analyse claim frequency using a Poisson or Negative Binomial model with treatment status as an explanatory variable and Exposure included appropriately. For an RCBD, the chosen blocking factor would also be included in the analysis. This would keep the statistical method consistent with the nature of the insurance response.

Summary

Experimental design could strengthen future investigations by allowing the insurer to test whether a safe-driving digital intervention causes changes in driving behaviour and claim frequency. CRD offers a simple randomized structure, while RCBD provides additional control over an important source of driver heterogeneity. Given the strong age-related differences found in the existing portfolio, an age-blocked RCBD is a

suitable option for a future pilot, provided that each block contains enough participants and the additional operational complexity can be managed.

The current dataset itself remains observational and is not used to claim an experimental treatment effect. Instead, the results from the earlier tasks are used to identify the risk factors that would be important when designing a future trial.

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Task 7 – Critical Evaluation of Principal Component Analysis

Introduction

Principal Component Analysis (PCA) is a multivariate technique that transforms correlated quantitative variables into a smaller set of uncorrelated principal components. The first components are constructed to retain as much of the variation in the original variables as possible (Jolliffe and Cadima, 2016). In this consultancy project, PCA was explored rather than assumed to be necessary. A small empirical analysis was carried out to determine whether the main numeric insurance predictors contain enough shared structure for meaningful dimensionality reduction.

The purpose of this section is therefore to combine the exploratory PCA results with a critical evaluation of its potential value for claim-frequency modelling. The assessment considers the amount of dimensionality reduction achieved, the interpretability of the resulting components, possible modelling benefits, and the situations in which PCA would or would not be appropriate for this dataset.

Exploratory PCA Method

Five numeric predictor variables were selected: VehPower, VehAge, DrivAge, BonusMalus and Density. ClaimNb was excluded because it is the response variable, while Exposure was excluded because it represents time at risk and is more appropriately handled separately in claim-frequency models. Categorical predictors such as Area, VehBrand, VehGas and Region were not included because conventional PCA is designed for quantitative variables.

The selected variables were standardized before PCA because their measurement scales differ substantially. For example, Density has a much larger numerical scale than the other variables. Standardization gives each selected variable comparable influence on the PCA rather than allowing variables with large numerical variance to dominate the result.

Variable	Mean	Standard deviation
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VehPower	6.455	2.051
VehAge	7.044	5.666
DrivAge	45.499	14.137
BonusMalus	59.762	15.637
Density	1,792.422	3,958.647

Correlation Structure Before PCA

PCA is most useful when several numeric variables share substantial correlation, because this creates redundancy that can potentially be represented by fewer components. The correlation matrix showed that most relationships between the selected variables were weak. The clearest relationship was between DrivAge and BonusMalus, with a correlation of approximately -0.480. The remaining correlations were generally close to zero, with absolute values below approximately 0.10.

Variable pair	Correlation	Interpretation
DrivAge – BonusMalus	-0.480	Moderate negative relationship
VehAge – Density	-0.090	Weak negative relationship
VehAge – BonusMalus	0.080	Weak positive relationship
BonusMalus – Density	0.078	Weak positive relationship
VehPower – BonusMalus	-0.076	Weak negative relationship

This weak overall correlation structure already suggests that the five variables contain relatively distinct information. Therefore, a strong reduction to only one or two principal components would not be expected.

Explained Variance Results

Component	Explained variance	Cumulative variance
PC1	30.25%	30.25%
PC2	21.97%	52.23%
PC3	19.90%	72.13%
PC4	17.67%	89.80%
PC5	10.20%	100.00%

The first principal component explained only 30.25% of the total variance. The first two components together retained 52.23%, while three components retained 72.13%. Approximately 89.80% of the variance was retained only after four components were included. Therefore, retaining around 80% or more of the information would require four components from the original five variables. This represents only a small dimensionality reduction.



Figure 1. Scree plot showing the explained variance of the five principal components.

The scree plot should be included because it gives a direct visual view of how gradually the explained variance declines. A strong PCA solution would normally show a small number of dominant components followed by a clear flattening of the curve. In this analysis, the variance is distributed across several components rather than being concentrated heavily in the first one or two.

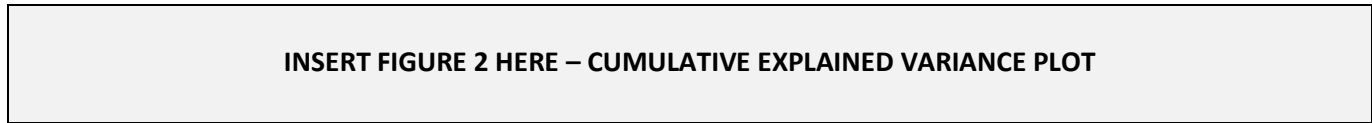


Figure 2. Cumulative explained variance as additional principal components are retained.

The cumulative variance plot is also useful because it makes the practical dimensionality decision clear. It shows that two components preserve only about half of the total variance and that four components are needed to approach 90%.

Principal Component Loadings

The component loadings show which original variables contribute most strongly to each principal component. The signs of PCA loadings can be reversed without changing the component, so interpretation should focus mainly on their relative magnitudes.

Variable	PC1	PC2	PC3	PC4	PC5
VehPower	-0.139	0.276	0.899	-0.300	0.078
VehAge	0.176	-0.599	0.433	0.648	-0.055
DrivAge	-0.680	-0.042	-0.068	0.250	0.685
BonusMalus	0.695	0.093	0.002	-0.044	0.712
Density	0.069	0.745	0.010	0.652	-0.123

PC1 is driven mainly by BonusMalus and DrivAge in opposite directions, which is consistent with their moderate negative correlation. PC2 is influenced mainly by Density and VehAge, while PC3 is dominated by VehPower. The fact that separate variables continue to dominate later components supports the conclusion that the numeric predictors are not highly redundant.

Is Dimensionality Reduction Necessary?

The empirical evidence suggests that dimensionality reduction is not necessary for the primary analysis. Only five numeric predictors were considered, and four principal components are needed to retain approximately 90% of their variance. Reducing five interpretable variables to four less interpretable components would provide little practical simplification.

The full insurance dataset also contains important categorical predictors that are not represented by this conventional PCA. Consequently, PCA would not reduce the entire modelling problem to a small set of components; it would only transform a subset of the available predictors.

Possible Advantages of PCA

PCA can still offer several potential benefits. It creates orthogonal components and can therefore reduce multicollinearity when predictors are strongly correlated. It can also reduce computational complexity and provide a compact representation of large groups of related numeric variables. Jolliffe and Cadima (2016) describe PCA as a method that can reduce dimensionality while attempting to minimize information loss, and Abdi and Williams (2010) similarly emphasize its value for extracting important information from sets of inter-correlated quantitative variables.

If a future version of the insurer's dataset contained many telematics, sensor or behavioural variables with strong correlation, PCA could become substantially more useful. In such a setting, many highly related measurements could potentially be compressed into a smaller number of components before predictive modelling.

Limitations for the Current Dataset

The main limitation is loss of interpretability. Original predictors such as DrivAge, VehAge and BonusMalus have direct business meaning for underwriting and risk management, while a principal component is a weighted combination of several variables. Replacing original variables with components would make model results harder to explain to management and operational users.

A second limitation is that PCA is unsupervised. The components are selected to explain variation in the predictor variables, not to maximize their relationship with ClaimNb. A component explaining relatively little predictor variance may still contain information that is useful for prediction. Jolliffe (1982) specifically showed that low-variance principal components can still be important in regression, meaning that components should not be discarded solely because they explain less variance.

The weak correlations observed in this dataset also reduce the expected benefit. PCA is most attractive when many predictors carry overlapping information. Here, most numeric predictors are only weakly correlated, so the analysis does not reveal a strong low-dimensional structure.

Expected Impact on Predictive Modelling

For the current claim-frequency problem, PCA is unlikely to provide a major improvement to the main predictive model. The modest reduction from five numeric predictors to four components would save little model complexity, while making the fitted model harder to interpret. It may also weaken business communication because management would need to understand component scores rather than direct effects associated with familiar insurance risk variables.

PCA could nevertheless be tested as an alternative modelling pipeline and its predictive performance compared with a model using the original predictors. If a future model suffered from serious multicollinearity or included a much larger number of correlated continuous variables, principal component regression could become more attractive. Any such comparison should be based on out-of-sample predictive performance rather than explained variance alone.

When PCA Should and Should Not Be Used

PCA may be useful when	PCA may be less suitable when
There are many correlated numeric predictors.	The number of numeric predictors is already small.
Dimensionality reduction is needed for computational or modelling reasons.	Most predictors are weakly correlated.
Multicollinearity is causing model instability.	Direct interpretation of individual risk factors is important.
A large set of telematics or sensor measures contains substantial redundancy.	Important predictors are categorical or have different data types.
Prediction is more important than explaining individual original variables.	Components are being selected only by variance without validating predictive value.

Overall Evaluation and Recommendation

The exploratory PCA provides evidence that PCA is technically possible but offers limited practical benefit for the present dataset. The five numeric predictors do not show strong overall correlation, and four principal components are required to retain about 89.8% of their total variance. This means the reduction in dimension is small.

For the main insurance claim-frequency model, retaining the original predictors is therefore preferable because it preserves interpretability and allows the insurer to relate model findings directly to driver, vehicle and geographic risk characteristics. PCA may be retained as an optional exploratory or comparison technique rather than the primary representation of the current data.

The recommendation could change if future data collection introduces a large number of correlated continuous variables, especially telematics or detailed behavioural measurements. In that situation PCA could provide greater reduction, reduce redundancy and make high-dimensional modelling more manageable.

Summary

PCA was evaluated using five standardized numeric predictors. PC1 explained 30.25% of total variance, the first two components explained 52.23%, and four components were required to retain 89.80%. The correlation matrix and loading structure show that most predictors contain relatively distinct information, with the strongest relationship occurring between DrivAge and BonusMalus.

These results indicate that dimensionality reduction is not currently necessary. PCA has recognized advantages for highly correlated and high-dimensional numeric data, but in this project the small reduction in dimension and loss of direct insurance interpretation outweigh the likely benefit. The original predictors should therefore remain the main inputs for predictive modelling, with PCA used only as a supplementary analytical option.

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Task 8 – Critical Evaluation of Bayesian Statistical Methods

Motor Insurance Claim-Frequency Consultancy

Updated with a practical Naïve Bayes and Bayesian Poisson demonstration

8.1 Introduction

Bayesian statistical methods provide an alternative framework for insurance risk analysis by combining observed data with prior assumptions and representing uncertainty through posterior distributions. For this consultancy, Bayesian methods are relevant because the response is a claim count, most policies record no claims, and the Task 5 Poisson model showed overdispersion. The assignment primarily requires a critical evaluation of Bayesian methods; the coding presented here is therefore used as supplementary evidence rather than as a replacement for the main Task 5 modelling exercise.

Three areas are considered: Naïve Bayes, Bayesian regression and Bayesian decision making. A small implementation was added to test how these ideas behave with the selected motor-insurance data. The Bayesian regression demonstration used a random sample of 5,000 policies and four standardized predictors (vehicle age, driver age, Bonus-Malus and log density), while retaining log(Exposure) as an offset.

8.2 Naïve Bayes

Naïve Bayes is mainly a classification method. To demonstrate it, ClaimNb was converted to a binary outcome: 0 for no claim and 1 for at least one claim. Gaussian Naïve Bayes was fitted using vehicle age, driver age, Bonus-Malus and log density.

Metric	Result
Overall accuracy	0.9475
No-claim recall	0.9970
Claim recall	0.0110
Claim precision	0.1623
Claim F1-score	0.0206
Test observations	135,603

Confusion matrix:

Actual / Predicted	No claim	Claim
No claim	128,404	387
Claim	6,737	75

The high accuracy is misleading because the data are strongly imbalanced. The classifier correctly identifies almost all no-claim policies, but it detects only about 1.1% of actual claim cases. Therefore, accuracy alone would create a false impression of strong performance. In addition, converting ClaimNb to a binary variable discards information about multiple claims and does not naturally preserve the exposure-adjusted claim-frequency structure.

Assessment: Naïve Bayes is useful as a simple secondary claim-occurrence classifier, but the practical results confirm that it is not an appropriate primary model for this consultancy.

8.3 Bayesian Regression

Bayesian regression is more directly aligned with the business problem. A Bayesian Poisson regression was fitted to claim counts with $\log(\text{Exposure})$ included as an offset. Normal priors were used for the intercept and regression coefficients. The four continuous predictors were standardized before modelling so that the coefficient magnitudes could be interpreted on a common scale.

8.3.1 Sampling diagnostics

The model was estimated with two MCMC chains, 500 posterior draws per chain after tuning, and no reported divergences. R-hat values ranged from 1.00 to 1.01 and effective sample sizes were generally high, indicating acceptable convergence for this small demonstration.

Parameter	Posterior mean	Lower 95%	Upper 95%	R-hat
Intercept	-2.348	-2.50	-2.20	1.00
Vehicle age	-0.401	-0.543	-0.254	1.01
Driver age	0.050	-0.084	0.169	1.00
Bonus-Malus	0.426	0.320	0.523	1.00
Log density	0.053	-0.064	0.177	1.01

Vehicle age has a clearly negative posterior relationship with claim frequency because its 95% credible interval is fully below zero. Since the predictor was standardized, the coefficient of approximately -0.401 implies that a one-standard-deviation increase in vehicle age is associated with an expected claim-frequency multiplier of $\exp(-0.401) \approx 0.67$, or roughly a 33% reduction, holding the other variables fixed within this simplified model.

Bonus-Malus has a clearly positive relationship with claim frequency. Its posterior mean is approximately 0.426 and the 95% credible interval remains above zero. On the standardized scale, a one-standard-deviation increase corresponds to a multiplier of $\exp(0.426) \approx 1.53$, or about a 53% increase in expected claim frequency, holding the other included variables fixed.

The credible intervals for driver age and log density include zero. Their directions are therefore less certain in this simplified Bayesian model. This should not be treated as evidence that these factors have no relationship with claims; the demonstration uses only 5,000 observations and a reduced set of predictors, while earlier tasks indicated that age and geography may have nonlinear or overlapping effects.



Figure 8.1. Posterior means and manually calculated 95% credible intervals for the Bayesian Poisson coefficients.

8.3.2 Posterior predictive check

Quantity	Value
Observed mean claim count	0.05320
Posterior predictive mean claim count	0.05285
Difference	-0.00035

The posterior predictive mean is very close to the observed mean claim count in the 5,000-policy sample. This indicates that the model reproduces the overall claim level reasonably well. However, this is only a basic posterior predictive check; a production model would also examine the full count distribution, zero frequency, calibration by risk group and out-of-sample performance.

8.3.3 Evidence from published research

Published insurance research supports the use of Bayesian methods for claim-frequency modelling. Zhang et al. (2024) developed Bayesian CART models with Poisson, Negative Binomial and zero-inflated Poisson distributions for insurance claim frequency. Tajuddin and Ismail (2022) compared frequentist and Bayesian zero-inflated models using Malaysian motor-insurance data and reported stronger fitting results for their Bayesian specifications in that dataset. Scheel et al. (2013) demonstrated the value of hierarchical Bayesian modelling for insurance claims with spatial structure, while Shams and Ghosh (2026) showed how Bayesian nonparametric regression can provide more flexible insurance claim modelling than a single standard Poisson relationship.

8.4 Bayesian Decision Making

Bayesian decision making goes beyond point prediction by using posterior probabilities to support an action. To demonstrate this idea, one policy was evaluated against the portfolio-average claim frequency.

Decision quantity	Result
Portfolio claim frequency	0.1021
Example policy posterior mean frequency	0.1167
$P(\text{policy risk} > \text{portfolio average})$	0.748
Illustrative review threshold	0.80
Illustrative decision	No additional underwriting review flag

Although the example policy has a posterior mean risk above the portfolio average, the posterior probability of being above average is about 74.8%. Under the illustrative 80% rule, the policy is not flagged. This demonstrates an important Bayesian distinction: a decision can depend on both the expected risk and the uncertainty surrounding that risk.

The 80% threshold is not proposed as an operational insurance rule. A real organization would select decision thresholds using the financial cost of manual review, consequences of false positives and false negatives, risk appetite, regulation and expert judgement. The current consultancy also models claim frequency only, so the result should support underwriting review or monitoring rather than automatically determine a premium.

8.5 Potential Advantages

Posterior distributions provide direct probability statements about parameter and policy risk uncertainty.

Prior actuarial information can be incorporated when it is relevant and defensible.

Hierarchical priors can partially pool information across sparse regions, vehicle brands or other categories.

Bayesian Poisson and Negative Binomial models can preserve the count-data and exposure structure of the insurance problem.

Posterior distributions can be updated as new claims experience becomes available.

Posterior probabilities can be linked directly to decision-support rules rather than relying only on point estimates.

8.6 Limitations and Implementation Challenges

Prior specification requires judgement and should be justified transparently.

Bayesian MCMC is more computationally demanding than a standard GLM, especially for the full portfolio of more than 678,000 policies.

Convergence, effective sample size, divergences and posterior predictive adequacy must be checked.

Complex Bayesian models may be more difficult to communicate and maintain operationally.

The practical demonstration is intentionally simplified: it uses 5,000 sampled policies and four continuous predictors, so it is not a direct performance comparison with the full Task 5 model.

Bayesian modelling cannot compensate for important business information that is absent from the dataset, such as claim severity and other commercial pricing factors.

8.7 When Bayesian Methods May Outperform Traditional Approaches

Bayesian methods are most likely to add meaningful value when:

some risk categories have limited observations and partial pooling can stabilize their estimates;

credible prior actuarial knowledge is available for a new or changing portfolio;

the uncertainty of a prediction is itself important to the decision;

overdispersion, heterogeneity or hierarchical geographical structure is difficult to represent with a simple Poisson GLM;

the insurer wants to update risk estimates sequentially as new claims information arrives; management decisions can be expressed in terms of posterior probabilities and decision costs.

8.8 Comparison with the Current Task 5 Models

Approach	Main strength	Main limitation / evidence from this project
Task 5 Poisson GLM	Interpretable, fast, good calibration and best successful test deviance	Overdispersion remained (Pearson dispersion ≈ 2.68).
LASSO / Elastic Net Poisson	Regularization and coefficient shrinkage	Did not improve test deviance over the full Poisson GLM.
Naïve Bayes demo	Very fast binary classification	94.75% accuracy but only 1.1% recall for the claim class; unsuitable as primary frequency model.
Bayesian Poisson demo	Posterior uncertainty and probability-based interpretation	Promising but fitted only to a 5,000-row simplified sample; not a full out-of-sample comparison.
Bayesian hierarchical / NB extension	Could address sparse groups and overdispersion more flexibly	More complex and computationally expensive; would require additional implementation and validation.

8.9 Overall Evaluation and Recommendation

The practical work strengthens the earlier critical evaluation. Naïve Bayes is not recommended as the main method because the apparently high accuracy is driven by the dominant no-claim class and the model detects very few claim cases. Bayesian Poisson regression is substantially more appropriate because it retains the count-data structure, accounts for exposure and provides posterior uncertainty.

The Bayesian demonstration produced stable MCMC diagnostics and sensible posterior patterns: vehicle age showed a negative relationship with claim frequency, while Bonus-Malus showed a positive relationship. The posterior predictive mean closely matched the observed mean in the demonstration sample. The decision example also showed how posterior probabilities can support underwriting review without turning the model into an autonomous pricing system.

For the current consultancy, the full frequentist Poisson GLM should remain the operational baseline because it has already been tested on the complete train/test split, is easy to interpret and is computationally efficient. The most meaningful Bayesian future extension would be a Bayesian Negative Binomial or hierarchical count model fitted to a broader predictor set. Such a model could address overdispersion, stabilize sparse categorical effects and provide richer uncertainty for decision support. Any production adoption should be based on a direct out-of-sample comparison with the Task 5 models rather than on the small demonstration alone.

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Task 9 – Time Series Analysis

9.1 Potential Application of Time Series Analysis

The current study uses the freMTPL2freq dataset to investigate motor-insurance claim frequency. The dataset contains 678,013 policy observations and includes variables such as claim count (ClaimNb), exposure (Exposure), driver age, vehicle age, bonus-malus score, vehicle characteristics, area and region. The analysis performed in the earlier tasks focuses mainly on explaining and predicting differences in claim frequency between policies rather than changes in claims through chronological time.

Therefore, a conventional time-series analysis cannot appropriately be performed using the current dataset alone. A time-series approach would require historical observations recorded in a consistent chronological sequence, such as monthly claim counts and corresponding monthly policy exposure over several years.

If such historical information were available, time-series analysis could complement the existing claim-frequency models by examining how the insurer's overall claims experience changes over time. This is relevant because forecasting claims can support insurance pricing, financial planning, risk management and operational planning.

9.2 Trend Analysis

The first application would be trend analysis, which identifies long-term increases or decreases in claim frequency. For example, the insurer could calculate monthly claim frequency as:

$$\text{Claim Frequency}_t = \text{Total Claims}_t / \text{Total Exposure}_t$$

where t represents a particular month.

Using exposure is important because an increase in the number of claims does not necessarily indicate increasing risk. The insurer may simply have insured more vehicles. Analysing claims relative to exposure would therefore provide a more meaningful measure of changes in underlying claim risk.

A time-series plot of monthly claim frequency could reveal whether claim frequency has been gradually increasing, decreasing or remaining relatively stable. Long-term changes could potentially be associated with changes in the insurer's portfolio, driving behaviour, road conditions, economic circumstances or other external factors.

This would extend the current cross-sectional analysis. The earlier models investigate which policy characteristics are associated with different claim frequencies, whereas trend analysis would examine how the insurer's overall claim frequency changes over time. These two forms of analysis therefore provide complementary information.

9.3 Seasonal Analysis

Historical monthly or quarterly data could also be analysed for seasonality. Seasonality refers to patterns that repeatedly occur at similar points within a fixed time period.

For example, claim frequency might systematically differ between months because of recurring changes in traffic volume, weather, holidays or travel behaviour. However, these patterns should be identified from the insurer's historical observations rather than assumed beforehand.

The time series could be decomposed conceptually as:

$$Y_t = T_t + S_t + E_t$$

where Y_t is the observed claim frequency, T_t is the long-term trend, S_t is the seasonal component, and E_t represents the unexplained or random component.

This decomposition would help distinguish a genuine long-term movement in claims from temporary recurring fluctuations. Recent non-life insurance research has also examined monthly insurance indicators using seasonal

forecasting methods such as SARIMA and exponential smoothing, demonstrating the relevance of explicitly considering seasonal structures in insurance data.

9.4 Forecasting Future Claim Frequency

Once temporal patterns were understood, the insurer could use historical observations to forecast future claim frequency. For example, if monthly data were available from several previous years, a model could use those observations to forecast claim frequency for subsequent months. Forecasts should include uncertainty rather than being treated as exact future values.

A possible time-series dataset could be structured as follows. The values shown are illustrative only and are not results from `freMTPL2freq`.

Month	Claims	Exposure	Claim Frequency
Jan 2023	1,250	18,500	0.0676
Feb 2023	1,180	18,200	0.0648
Mar 2023	1,340	18,900	0.0709

The time-series model would learn from previous observations and produce forecasts for later periods. Forecast accuracy should be assessed using a genuine temporal holdout or rolling-origin evaluation rather than randomly splitting observations, because random splitting can allow future observations to influence evaluation of a model intended to predict the future.

9.5 ARIMA Modelling

One appropriate statistical approach would be the Autoregressive Integrated Moving Average (ARIMA) model.

$$ARIMA(p, d, q)$$

In this notation, p represents the autoregressive component and describes relationships with previous observations; d represents differencing used to address non-stationarity; and q represents the moving-average component, representing dependence on previous forecast errors.

For example, current claim frequency may partly depend on recent claim-frequency observations. If the series shows a long-term trend, differencing could be considered to obtain a more stationary series before modelling.

Before fitting an ARIMA model, the insurer would examine the chronological series, investigate stationarity and inspect autocorrelation patterns. ACF and PACF plots could help identify possible autoregressive and moving-average structures. Candidate models could then be compared using criteria such as AIC together with out-of-sample forecasting measures such as MAE and RMSE.

Where strong seasonal behaviour exists, SARIMA (Seasonal ARIMA) would be more appropriate because it extends ARIMA with seasonal components. Recent insurance forecasting research demonstrates that ARIMA-family models remain useful forecasting benchmarks. However, model selection should depend on validated forecasting performance rather than assuming that ARIMA is automatically the best approach.

9.6 Critical Evaluation of ARIMA

ARIMA would have several advantages for the insurer. It is an established statistical method, is relatively interpretable and can model temporal dependence that the existing cross-sectional claim-frequency models do not directly address.

However, ARIMA also has limitations. It mainly relies on historical patterns in the series and may perform poorly when relationships become strongly nonlinear or when sudden structural changes occur. Forecasting performance can also deteriorate when the future behaves substantially differently from the historical period used to estimate the model.

Recent insurance research highlights the importance of accurate claim forecasting for pricing, profitability, business planning and risk management. At the same time, greater model complexity does not automatically guarantee better forecasts. Therefore, ARIMA or SARIMA should be treated as candidate models and compared

with appropriate baseline and alternative forecasting approaches using out-of-sample performance, interpretability and operational requirements.

9.7 Relationship with the Existing Statistical Models

Time-series analysis would not replace the Poisson and Negative Binomial claim-frequency modelling considered earlier in this project. Instead, the approaches answer different questions.

Policy characteristics → Expected claim frequency

The existing statistical models allow investigation of whether policy-level characteristics such as driver age, vehicle age, bonus-malus score, fuel type and geographical characteristics are associated with claim frequency.

Historical claim behaviour → Future claim behaviour

A time-series model would instead examine how aggregate claim behaviour evolves over time and use past temporal patterns to forecast future behaviour. Combining the approaches could therefore provide the insurer with a more complete analytical system: policy-level models could support risk differentiation, while time-series models could support portfolio-level forecasting and planning.

9.8 Possible Business Applications

If reliable historical data were collected, time-series forecasts could support several organizational activities. Forecasts of future claim frequency could help the insurer anticipate periods of increased claims and allocate claims-handling resources accordingly. They could also contribute to budgeting, financial planning, reserve-related analysis and monitoring changes in portfolio risk.

Forecasts could additionally provide an early-warning function. If actual claim frequency began consistently exceeding expected levels, management could investigate whether the difference resulted from changes in the insured portfolio, external conditions or previously unidentified risk factors.

9.9 Expected Organizational Benefits

Introducing a time-series component could improve the insurer's ability to move from analysing historical policy-level relationships toward forward-looking decision support.

The main benefits would include improved planning for expected claim volumes, better allocation of operational resources, earlier identification of changes in claims experience and additional evidence for financial and risk-management decisions. Historical forecasts could also be updated as new monthly information becomes available, creating an ongoing monitoring system rather than a one-time statistical analysis.

However, these benefits depend strongly on data quality and temporal coverage. A sufficiently long and consistently recorded historical series would be necessary to distinguish meaningful trends and seasonality from random fluctuations. Structural changes could also make historical patterns less representative of future experience.

9.10 Conclusion

Time-series analysis would be a valuable future extension of the current motor-insurance study if historical time-dependent data became available. Trend analysis could identify long-term changes in claim frequency, while seasonal analysis could reveal recurring patterns. ARIMA or SARIMA could then provide interpretable statistical forecasts based on temporal dependencies.

This analysis should complement rather than replace the Poisson and Negative Binomial claim-frequency modelling considered earlier in the project. The existing models explain differences in risk between policies, whereas time-series models would investigate how aggregate claim behaviour evolves through time.

For the insurer, combining these perspectives could provide stronger support for claims planning, risk monitoring, budgeting and other forward-looking decisions. Nevertheless, ARIMA should be treated as a

candidate rather than an automatically preferred model, with forecasting approaches compared using appropriate out-of-sample temporal validation.

References

Perla, F., Richman, R., Scognamiglio, S. and Wüthrich, M.V. (2023). Machine Learning in Forecasting Motor Insurance Claims. *Risks*, 11(9), 164. <https://doi.org/10.3390/risks11090164>

Peovski, F. et al. (2024). Comparative analysis of forecasting models in the nonlife insurance: Insights from the SARIMA and ETS approaches. *Risk Management and Insurance Review*. <https://doi.org/10.1111/rmir.12288>

Task 10 – Industry Innovation Proposal

Introduction

The earlier analysis identified clear differences in motor insurance claim frequency across several policyholder, vehicle and geographical characteristics. Driver age, vehicle age, BonusMalus, Area and population density were particularly useful for understanding how claim risk varies across the portfolio. However, statistical findings create value only when they are translated into a practical decision process that can support day-to-day insurance operations.

Based on these findings, the proposed innovation is a Motor Insurance Risk Intelligence and Decision Support Framework. The system would combine historical policy and claim data, statistical modelling and an interactive management dashboard to estimate claim frequency risk, classify policies into understandable risk groups, explain the main contributing factors and support underwriting, portfolio monitoring and risk-management decisions.

Business Need

Motor insurers manage large portfolios containing drivers and vehicles with very different levels of risk. Traditional manual review can be inconsistent and may not make full use of the patterns available in historical data. The organization therefore needs a structured way to identify higher-risk policies earlier, monitor changing portfolio risk and provide decision-makers with evidence that is easy to understand.

The project findings show why such a system is useful. Younger drivers, newer vehicles, higher BonusMalus groups and higher-density areas displayed materially different claim-frequency patterns. A decision-support framework can combine these signals rather than requiring staff to interpret each variable separately.

Proposed Solution

The proposed solution is a centralized risk-intelligence framework that converts policy-level information into a claim-frequency risk estimate and a practical decision view. It is intended to support human insurance professionals rather than automatically replace underwriting or pricing judgement.

System component	Purpose
Data input layer	Collect policyholder, vehicle, exposure, geographical and historical claim information.
Data quality layer	Check missing values, invalid ranges, duplicates and inconsistent records before scoring.
Risk model	Use the validated claim-frequency model selected in Task 5 to estimate expected claim risk.
Risk segmentation	Translate model output into clear Low, Medium and High risk categories.
Explainability layer	Show the main characteristics associated with the policy's estimated risk.
Decision-support dashboard	Provide policy-level and portfolio-level summaries, trends and alerts.
Monitoring layer	Compare predicted risk with later claim outcomes and identify model drift or changing risk patterns.

Conceptual Framework

The proposed framework follows a continuous decision-support cycle. New policy information is validated, transformed into analytical inputs and passed through the claim-frequency model. The estimated risk is then converted into a form that operational teams can use. Actual future claim outcomes are returned to the monitoring process so that the organization can evaluate whether the model remains reliable.

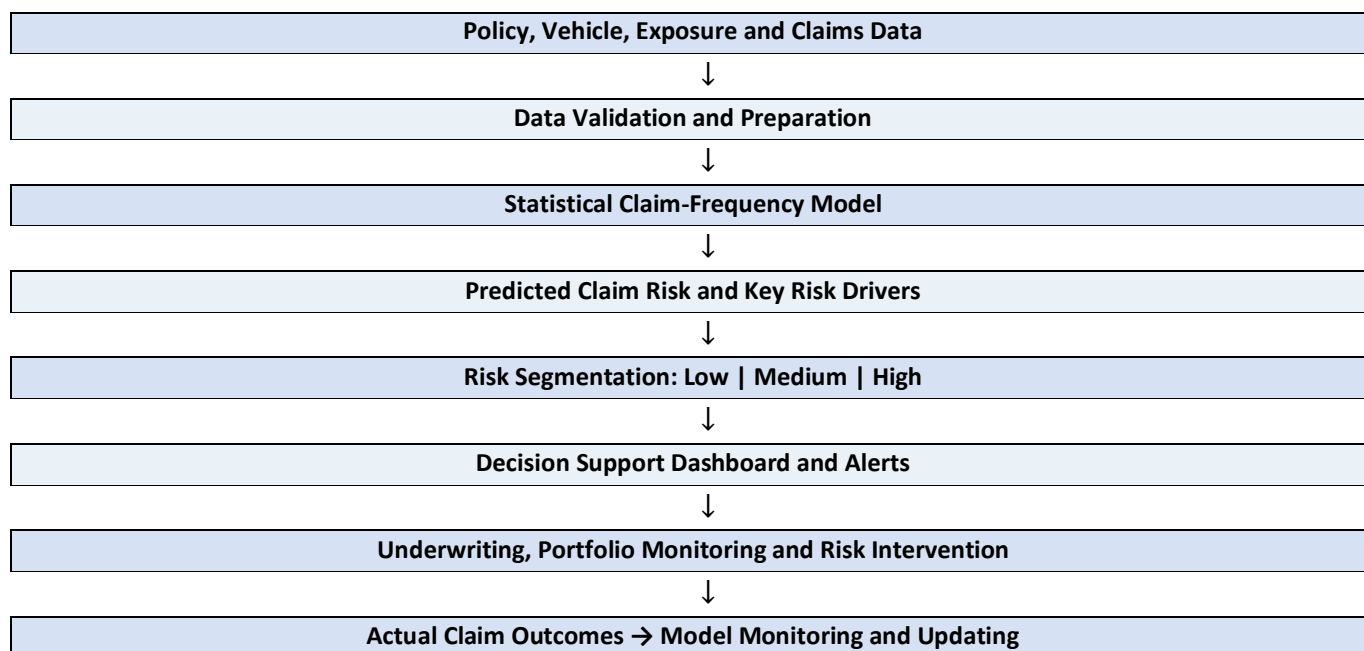


Figure 1. Conceptual framework for the Motor Insurance Risk Intelligence and Decision Support Framework.

Main System Outputs

At policy level, the system would present the estimated claim-frequency risk together with a simple risk category and the main factors associated with that estimate. At portfolio level, management would be able to view the concentration of higher-risk policies across age groups, vehicle groups, BonusMalus categories, Areas and other relevant segments.

Output	Example use
Predicted claim frequency	Provides a quantitative estimate of expected claim risk.
Risk category	Simplifies interpretation by grouping policies into operational risk bands.
Key risk drivers	Shows which characteristics are contributing most strongly to the estimated risk.
Segment summaries	Shows how risk is distributed across age, vehicle, geography and BonusMalus groups.
Risk alerts	Highlights unusually high-risk policies or portfolio segments for further review.
Model monitoring indicators	Tracks whether prediction quality changes as new claims are observed.

Organizational Use

The framework is designed as a decision-support tool rather than an automatic decision engine. High-risk classification should trigger additional review, not an automatic rejection or price change. This keeps professional judgement within the process and reduces the risk of inappropriate decisions based on a single statistical score.

Business area	How the framework could support decisions
Underwriting	Identify policies that may require additional review before acceptance or renewal.
Risk management	Monitor segments with consistently elevated claim frequency and design targeted interventions.

Actuarial / pricing	Provide additional empirical evidence for investigating risk differentiation and pricing assumptions.
Management	View portfolio risk composition and monitor whether high-risk concentrations are changing.
Customer engagement	Support targeted safety or prevention programmes for suitable risk segments.

Expected Benefits

The main benefit is that the organization's existing data would be converted into a consistent and repeatable decision process. This could improve the speed and quality of risk assessment, make portfolio monitoring more proactive and reduce dependence on purely manual interpretation.

Expected benefit	Business value
Earlier risk identification	Higher-risk policies or segments can be reviewed before losses accumulate.
More consistent decisions	Staff use the same analytical evidence instead of relying only on individual judgement.
Better portfolio visibility	Management can identify where risk is concentrated across the book of business.
Explainable analytics	Risk estimates are accompanied by recognizable factors such as age, vehicle age and BonusMalus.
Continuous learning	New claim outcomes can be used to monitor and periodically update the analytical model.
Support for prevention	High-risk segments can be considered for targeted safety interventions rather than only reactive claims management.

Implementation Approach

The solution can be introduced gradually rather than deployed as a fully automated system from the beginning. A phased implementation would allow the insurer to validate the model, test operational workflows and obtain staff feedback before wider use.

Phase	Main activities
1. Data integration	Create a reliable analytical dataset from policy, exposure and claims systems.
2. Model validation	Validate the selected claim-frequency model on historical and holdout data.
3. Prototype dashboard	Build a simple dashboard showing risk scores, risk segments and major risk drivers.
4. Pilot deployment	Use the system with a limited underwriting or portfolio-management team.
5. Operational integration	Connect scoring and dashboard outputs to normal business workflows.
6. Ongoing monitoring	Track data quality, model performance, drift, fairness and user feedback.

Required Resources

The project does not require a completely new insurance information system, but it does require reliable access to existing policy and claims data, analytical skills and a platform for presenting the results. The exact technology can be selected according to the organization's existing infrastructure.

Resource	Requirement
Data	Historical policy, claim, exposure and relevant customer / vehicle data.
Analytics	Python or R environment for statistical modelling and validation.
Data platform	Database or data warehouse capable of supplying clean analytical data.
Dashboard	Business intelligence platform such as Power BI, Tableau or an equivalent internal tool.
People	Data analyst / statistician, data engineer, underwriter or actuarial expert, and IT support.
Governance	Processes for privacy, access control, model approval, monitoring and change management.

Implementation Challenges

The usefulness of the framework depends on the quality and stability of the underlying data and model. Poor-quality source data could produce misleading risk estimates. Integration with existing systems may also require technical work, especially if policy and claims data are stored in separate platforms.

Model drift is another important issue. Claim behaviour may change over time because of economic conditions, changes in vehicle technology, customer behaviour or underwriting practices. The model should therefore be monitored and periodically revalidated rather than treated as permanent.

Staff adoption and explainability must also be considered. A statistically accurate model will have limited organizational value if underwriters and managers do not understand or trust its outputs. The dashboard should therefore present both the risk estimate and the main contributing factors in simple business language.

Ethical and Risk Management Considerations

Insurance risk analytics can influence financially important decisions, so the framework should include clear governance. Personal information should be processed only for legitimate business purposes and protected through appropriate access controls. Model outputs should be reviewed for systematic unfairness across customer groups, and variables should not be used simply because they improve prediction if their use creates legal, ethical or reputational concerns.

Human oversight should remain part of high-impact decisions. The model should be treated as additional evidence for qualified staff rather than an automatic rule for policy rejection, pricing or customer treatment. Decisions and model versions should also be auditable so that the organization can explain how analytical information was used.

Expected Organizational Impact

The proposal would move the organization from mainly descriptive use of historical claims toward proactive risk intelligence. Instead of reviewing past claims only after they occur, management could use estimated future claim frequency to prioritize investigation, monitor risk concentrations and design prevention strategies.

It would also create a common analytical view across underwriting, risk management, actuarial and management teams. This could improve communication between departments because the same risk measures and segment definitions would be used when discussing portfolio risk.

Success Measures

The innovation should be evaluated using both statistical and operational measures. Strong predictive performance alone would not demonstrate that the system is useful in practice.

Measure	Purpose
Predictive performance	Check whether estimated claim risk remains accurate on new data.
Calibration by risk segment	Check whether higher predicted-risk groups actually experience higher claim frequency.
Data quality rate	Monitor missing, invalid or delayed input data.
User adoption	Measure whether intended staff actively use the dashboard in decision workflows.
Review efficiency	Assess whether the system helps staff prioritize cases more efficiently.
Fairness / governance indicators	Monitor whether model-supported decisions create unacceptable disparities or control failures.

Summary

The proposed Motor Insurance Risk Intelligence and Decision Support Framework converts the project's statistical findings into a practical organizational solution. It combines data validation, claim-frequency modelling, risk segmentation, explainable outputs, dashboard reporting and continuous monitoring in one decision-support process.

The framework could help the insurer identify higher-risk policies earlier, improve consistency in underwriting and portfolio monitoring, and support targeted risk-management actions. Its value depends on reliable data, model validation, explainability, human oversight and ongoing monitoring. The proposal therefore uses statistical prediction as decision support rather than as an uncontrolled automated decision system.

Task 12 – Final Consultancy Recommendations

Draft completion note: The statistical and research-based recommendations are completed. The Industry Expert Feedback fields must be replaced with genuine Task 11 feedback before final submission. The final Task 5 model name/performance should also be inserted once that modelling task is completed.

Introduction

This section converts the consultancy findings into practical recommendations for senior management. The recommendations are based on the descriptive and inferential evidence already obtained from the freMTPL2freq portfolio, the research landscape reviewed in Task 2, and the proposed Risk Intelligence and Decision Support Framework developed in Task 10.

The recommendations should not be interpreted as instructions for automatically changing insurance premiums. The project focuses on claim frequency only, while a full commercial pricing decision would also require claim severity, expenses, profitability, regulatory requirements and other business considerations. The proposed use of statistical modelling is therefore decision support for underwriting, portfolio management and risk monitoring.

Executive Recommendation Summary

Priority	Recommendation	Management purpose
1	Implement the Motor Insurance Risk Intelligence and Decision Support Framework	Convert statistical findings into a repeatable risk-monitoring and underwriting-support process.
2	Adopt multi-factor risk segmentation rather than single-variable rules	Use several validated risk indicators together when reviewing portfolio risk.
3	Strengthen monitoring and prevention for higher-risk segments	Direct additional review and preventive action toward segments with materially elevated claim frequency.
4	Establish model governance, fairness checks and human oversight	Control model risk and prevent inappropriate automated customer decisions.
5	Expand data collection and continue model research	Improve future modelling with richer behavioural, severity and time-dependent information.

Strategic Recommendation 1 – Implement the Risk Intelligence and Decision Support Framework

Senior management should progress the Task 10 proposal from a conceptual framework to a controlled internal pilot. The system should combine validated policy data, the final claim-frequency model from Task 5, risk segmentation, explainable outputs and a management dashboard. Its role should be to identify and monitor risk patterns and support professional judgement rather than automatically determine premiums or policy acceptance.

Statistical evidence	Task 4 found statistically significant differences in exposure-adjusted claim rates across driver age, vehicle age, Area, BonusMalus and Density (all overall Poisson deviance tests $p < 2.2 \times 10^{-16}$). These findings demonstrate that the portfolio contains identifiable risk structure that can be summarized in a decision-support system.
Research support	The Task 2 literature review concluded that interpretable GLMs remain practically valuable in insurance and that model selection should balance predictive performance, calibration, interpretability, assumptions, complexity and business usefulness. It also identified practical decision support as an important research-to-practice gap.

Industry expert feedback	TO BE COMPLETED AFTER TASK 11: Insert the expert's genuine comments on whether the proposed framework is practical, useful and feasible for insurance operations.
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Management action: begin with a limited pilot used by underwriting or portfolio-risk staff, measure whether the system improves review consistency and risk visibility, and expand only after validation and user feedback.

Strategic Recommendation 2 – Use Multi-Factor Risk Segmentation

The organization should avoid making risk decisions from a single variable. Instead, driver, vehicle, geographical and exposure-related information should be considered together through the final multivariable model and then summarized into understandable risk segments.

Statistical evidence	Several variables showed different claim patterns. Driver age was strongly associated with claim frequency: the 18–24 group had a frequency of about 0.189, while older groups were approximately 0.091–0.106. Vehicle age also differed substantially, from about 0.138 for vehicles aged 0–2 years to 0.040 for the 31+ group. Area F had an exposure-adjusted rate about 1.70 times Area A, and the High Density group about 1.49 times the Low Density group.
Research support	Task 2 found that no single modelling technique or isolated factor should be treated as universally sufficient. The literature supports multivariable count models because claim frequency depends on several risk characteristics and exposure, while interpretable statistical models allow individual relationships to be examined and communicated.
Industry expert feedback	TO BE COMPLETED AFTER TASK 11: Insert genuine expert feedback on whether multi-factor segmentation reflects real underwriting practice and which factors the expert considers operationally meaningful.

Management action: define risk bands only after the Task 5 model is finalized and calibrated. Risk bands should be reviewed periodically rather than treated as permanent categories.

Operational Recommendation – Strengthen Monitoring of Higher-Risk Segments

Operational teams should use the statistical evidence to prioritize additional monitoring, investigation and preventive interventions for segments with consistently elevated claim frequency. This should be a prioritization mechanism, not a rule for automatic rejection.

Statistical evidence	The BonusMalus analysis produced the strongest gradient among the grouped variables: relative to BonusMalus 50–59, the estimated claim-rate ratios increased to about 1.54 (60–69), 1.60 (70–79), 1.73 (80–99), 3.76 (100–149), 6.69 (150–200) and 30.71 (201+). The extreme groups are small and must therefore be interpreted cautiously. Younger drivers and high-density locations also showed elevated claim rates.
Research support	The Task 2 research landscape emphasizes that statistical models can support underwriting, portfolio monitoring and risk segmentation. It also notes that model outputs should be interpreted with attention to stability, practical usefulness and data limitations rather than predictive performance alone.
Industry expert feedback	TO BE COMPLETED AFTER TASK 11: Insert the expert's genuine view on whether these high-risk segments should trigger review, customer education, safety interventions or another practical action.

Management action: configure dashboard alerts for materially elevated predicted risk, then require a human reviewer to determine whether additional information or preventive action is appropriate.

Risk Management Recommendation – Establish Model Governance and Continuous Validation

The insurer should establish a formal model-governance process before statistical scores are used in routine operations. This should include data-quality checks, model-performance monitoring, calibration review, drift detection, documentation of model versions and periodic revalidation.

Statistical evidence	The current dataset contains 678,013 observations and is structurally complete, but ClaimNb is highly zero-dominated and the Poisson analyses rely on count-model assumptions. The project therefore treats Poisson as a baseline and recognizes the need to evaluate overdispersion, excess-zero structure and alternative count models before final deployment.
Research support	Task 2 explicitly recommends diagnostic, data-dependent model selection rather than automatically assuming Poisson is final. It identifies Negative Binomial and, where justified, zero-inflated models as alternatives. The literature review also notes that dynamic risk can change with the insured population, vehicle technology, regulation, economic conditions and driving behaviour.
Industry expert feedback	TO BE COMPLETED AFTER TASK 11: Insert the expert's genuine comments on model review, data quality, operational monitoring and how often an insurance risk model should be reassessed.

Management action: assign clear responsibility for model ownership, validation and approval. The production model should have documented thresholds for retraining or review if calibration or portfolio behaviour changes materially.

Ethical Recommendation – Maintain Fairness, Transparency and Human Oversight

Model-supported insurance decisions can affect customers financially. The organization should therefore require transparent use of model outputs, fairness monitoring and human review for high-impact decisions. A variable should not be used solely because it improves prediction if its use cannot be justified ethically, legally or operationally.

Statistical evidence	The statistical analysis identified strong differences across age, geographical and rating-related groups. These associations are useful for understanding portfolio risk but do not establish causal responsibility for individual customers. Very large sample sizes can also make small effects statistically significant, so business decisions must consider practical effect size as well as p-values.
Research support	Task 2 highlights fairness, transparency and explainability as central requirements for responsible insurance modelling. It notes that predictive accuracy alone is insufficient and that complex models may identify patterns that are difficult to justify from a business or ethical perspective.
Industry expert feedback	TO BE COMPLETED AFTER TASK 11: Insert the expert's genuine feedback on customer fairness, explainability, acceptable use of risk variables and the importance of professional judgement.

Management action: use model scores as additional evidence rather than automatic decisions; document the reasons for consequential actions and provide a process for review when model-supported decisions are challenged.

Future Research Opportunities

The current dataset supports a strong claim-frequency analysis but does not represent every component required for a complete insurance risk system. The following extensions are recommended for future work.

Future direction	Reason
Add claim severity and cost data	Allows frequency and severity to be combined into a fuller expected-loss framework.
Collect richer behavioural / telematics variables	May reveal driving behaviour that is not represented by static policy characteristics.
Obtain longitudinal and time-stamped claims data	Would support trend, seasonality, forecasting and dynamic risk updating.

Externally validate the model on a different portfolio or time period	Tests whether findings generalize beyond the current dataset.
Evaluate Bayesian models where uncertainty or prior knowledge is valuable	Bayesian regression can represent parameter uncertainty and incorporate credible prior actuarial information.
Test risk-reduction interventions experimentally	A randomized pilot could provide causal evidence on whether safe-driving interventions reduce risky behaviour or claims.
Statistical evidence	Task 7 found that PCA offered limited reduction for the current five numeric predictors: four components were required to retain about 89.8% of variance, so preserving original variables was preferred. Earlier tasks also identified the limitations of a static, claim-frequency-only dataset.
Research support	Task 2 identifies dynamic modelling, Bayesian approaches, machine learning and responsible risk modelling as relevant extensions when the available data and business need justify the additional complexity.
Industry expert feedback	TO BE COMPLETED AFTER TASK 11: Insert the expert's genuine priorities for additional data, future modelling or practical research.

Recommended Implementation Priorities

Time horizon	Priority action
Immediate	Complete Task 5 model comparison and validation; finalize Task 11 expert consultation; define governance requirements.
Short term	Build a small dashboard prototype using validated model outputs and existing portfolio data.
Pilot stage	Use the framework with a limited internal team and collect feedback on usefulness, interpretation and workflow impact.
Ongoing	Monitor calibration, data quality, fairness and model drift; review risk bands as portfolio behaviour changes.
Long term	Expand the data foundation with severity, behavioural and longitudinal information and evaluate more advanced approaches only where they add measurable value.

Final Management Recommendation

The central recommendation is to adopt an evidence-based motor-insurance risk analytics process that combines interpretable statistical modelling with professional judgement. The analysis demonstrates that claim frequency varies materially across several driver, vehicle and geographical characteristics, but these findings should be used as part of a multivariable and governed decision-support process rather than as isolated rules.

The proposed Risk Intelligence and Decision Support Framework provides a practical route for converting the statistical work into organizational value. Its implementation should be gradual, transparent and continuously monitored. More advanced modelling should be adopted only when it provides a clear improvement in statistical validity, predictive performance or business usefulness without creating unacceptable complexity or fairness risk.

Before final submission, the genuine Task 11 industry-expert feedback must be inserted into each recommendation so that the final section satisfies the assignment requirement that every recommendation is supported by statistical evidence, research literature and industry expert feedback.

